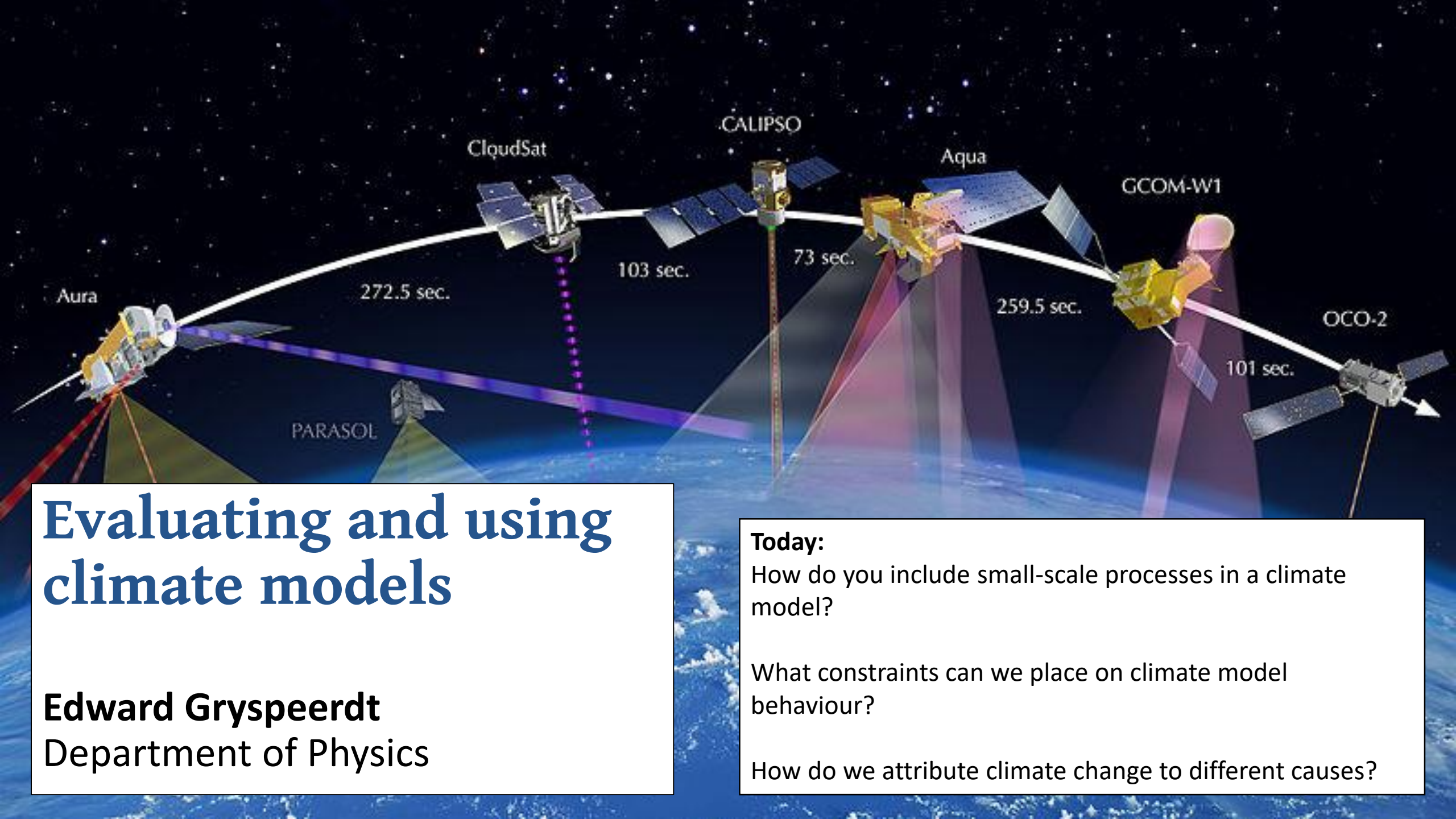




Evaluating and using climate models

Edward Gryspeerdt
Department of Physics

Today:

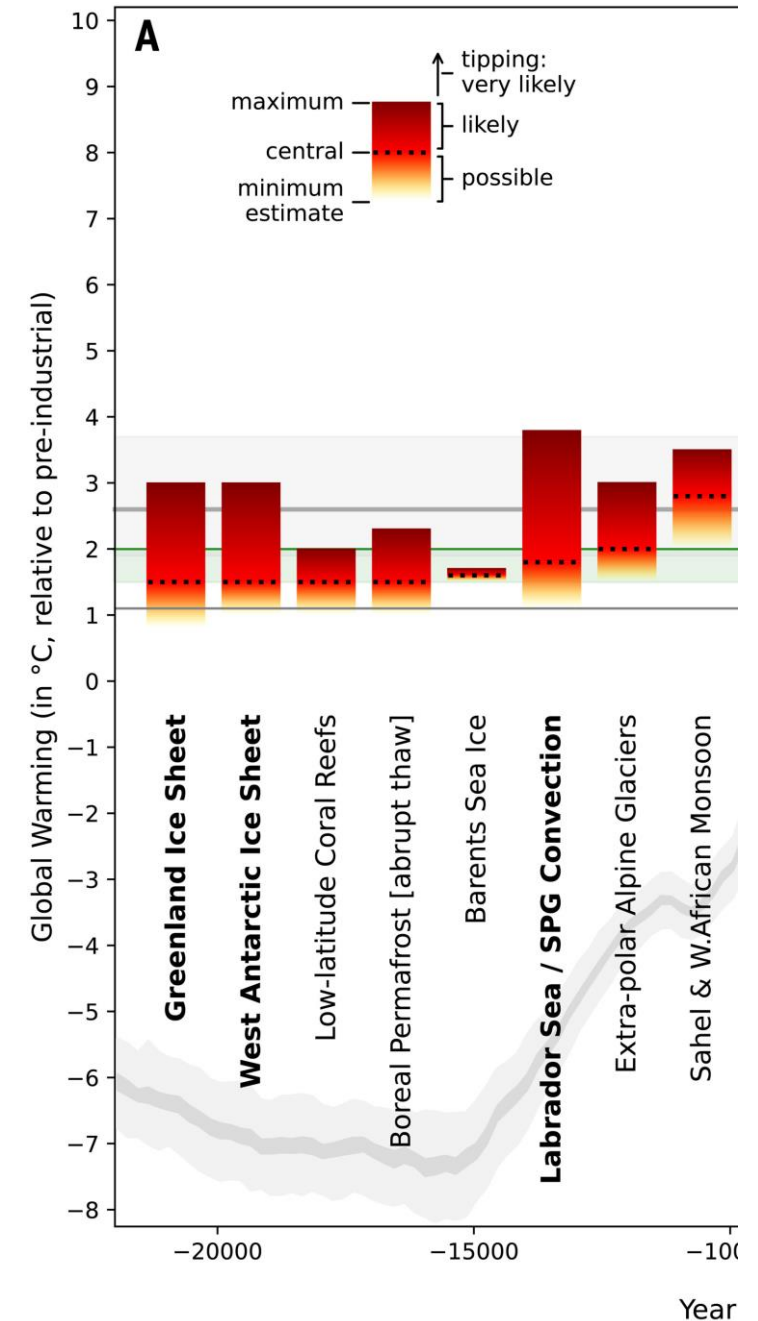
How do you include small-scale processes in a climate model?

What constraints can we place on climate model behaviour?

How do we attribute climate change to different causes?

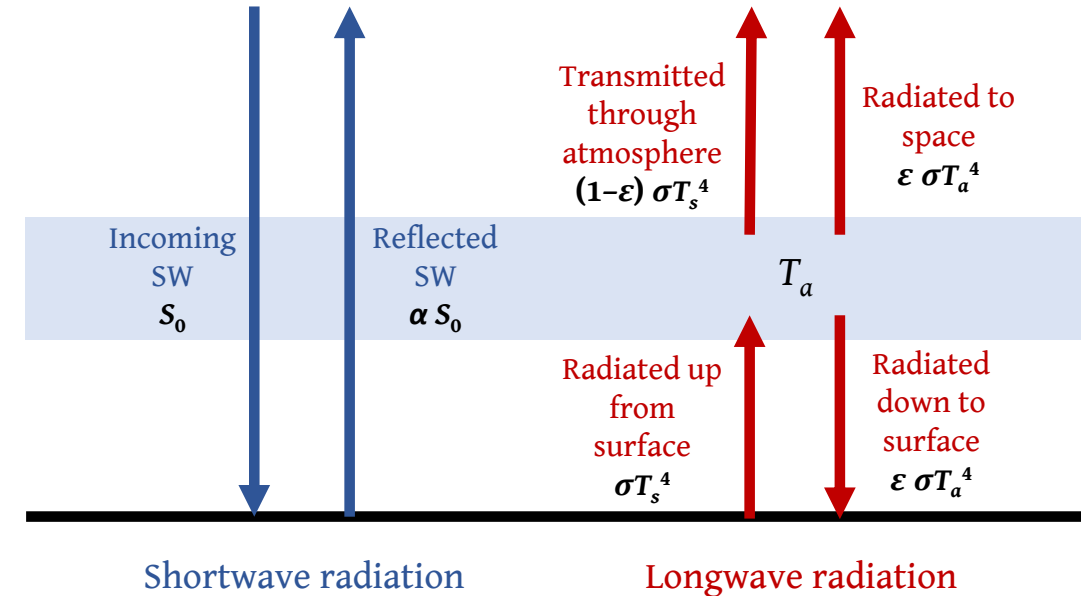
Recap: Tipping points

- Volcanoes inject large amounts of sulphur dioxide into the stratosphere
 - This creates a layer of aerosol that reflects sunlight, cooling the Earth
- The climate has a number of 'tipping points' where going past them can accelerate change
 - Returning to a temperature below the tipping point threshold does not restore the system
- The locations of tipping points are uncertain
 - They are the driver behind efforts to limit warming to 2 (or 1.5C)



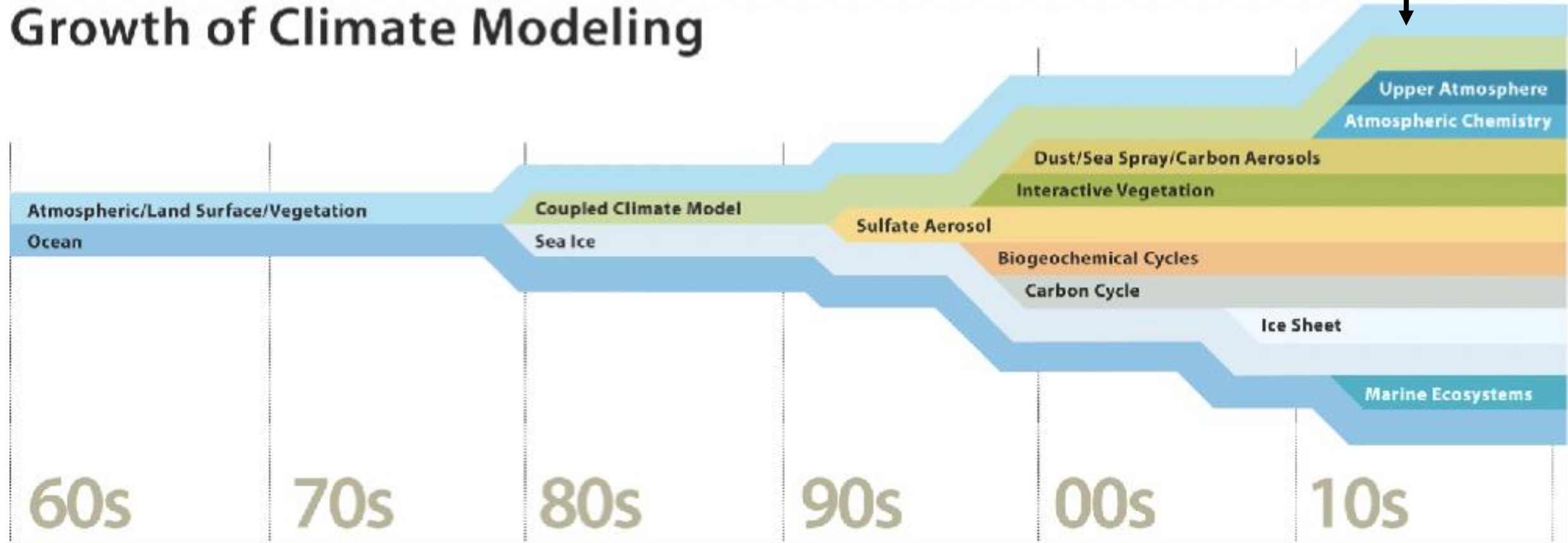
Recap: Simple climate models

- Climate models are a simplified representation of the world
- Weather and climate models are very similar, but solve different problems
 - Weather is an 'initial-value' problem
 - Climate is a 'boundary-value' problem
- A simple energy balance model requires an atmosphere to get a sensible surface temperature
 - A greenhouse effect is required!
 - A more complex simulation is needed to represent other aspects of the Earth system



Increasing complexity

Growth of Climate Modeling



Solving the equations

- For each grid box in atmosphere, ocean and ice, a set of equations need to be solved
- These equations almost always have a d/dt term on the left hand side: they provide the **rate of change** of variables such as temperature, wind, humidity, etc
- This allows the model to predict how these variables will evolve in time
- The model predicts the changes over short “**time steps**”, typically about 30 minutes, then repeats the prediction with the updated variable values

east-west wind

$$\frac{du}{dt} - \left(f + u \frac{\tan \phi}{a} \right) v = -\frac{1}{a \cos \phi} \frac{1}{\rho} \frac{\partial p}{\partial \lambda} + F_\lambda$$

$$\frac{dv}{dt} + \left(f + u \frac{\tan \phi}{a} \right) u = -\frac{1}{\rho a} \frac{\partial p}{\partial \phi} + F_\phi$$

$$g = -\frac{1}{\rho} \frac{\partial p}{\partial z}$$

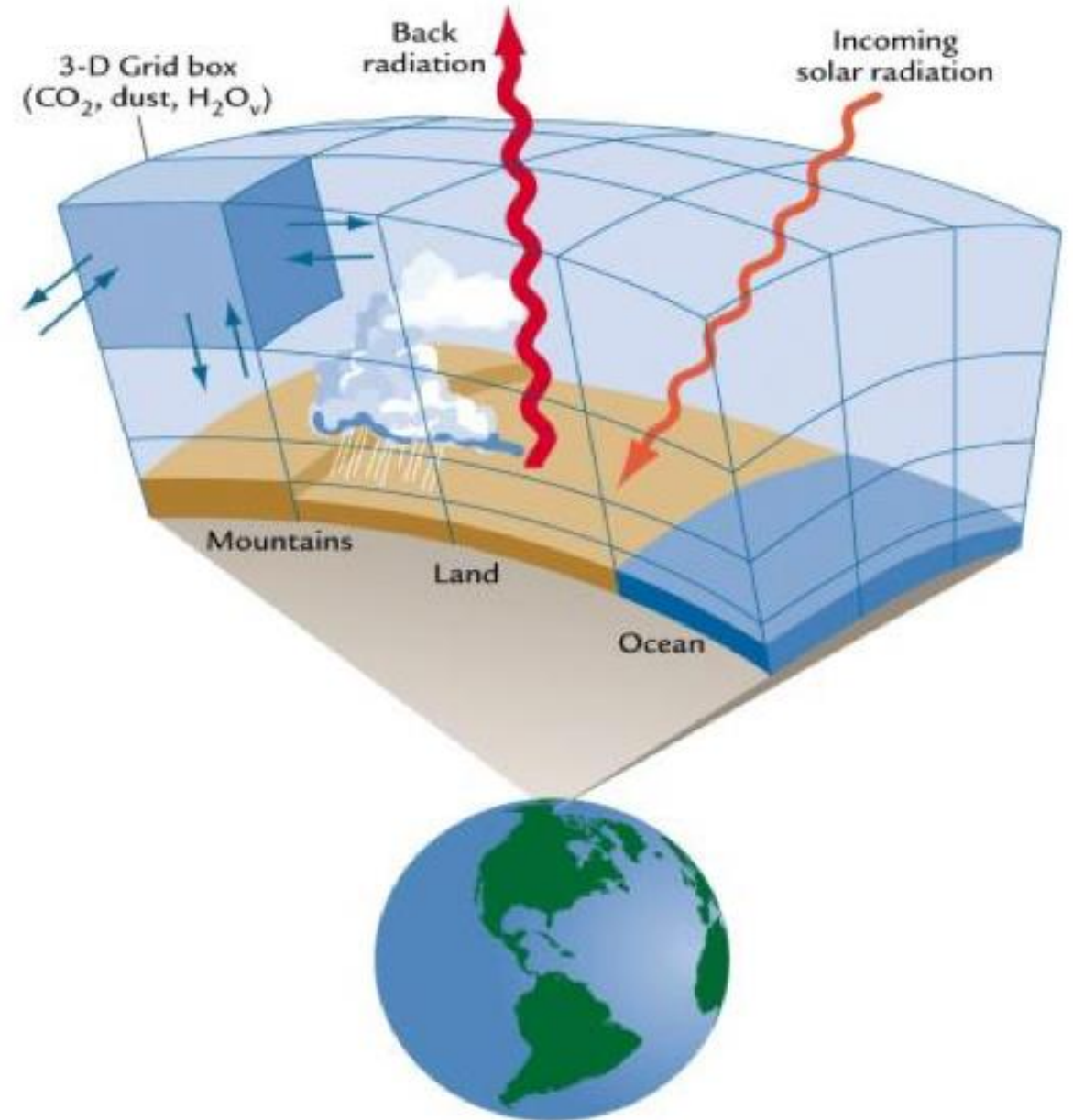
$$\frac{\partial \rho}{\partial t} = -\frac{1}{a \cos \phi} \left[\frac{\partial}{\partial \lambda} (\rho u) + \frac{\partial}{\partial \phi} (\rho v \cos \phi) \right] - \frac{\partial}{\partial z} (\rho w)$$

$$c_p \frac{dT}{dt} - \frac{1}{\rho} \frac{dp}{dt} = Q$$

$$p = \rho R T$$

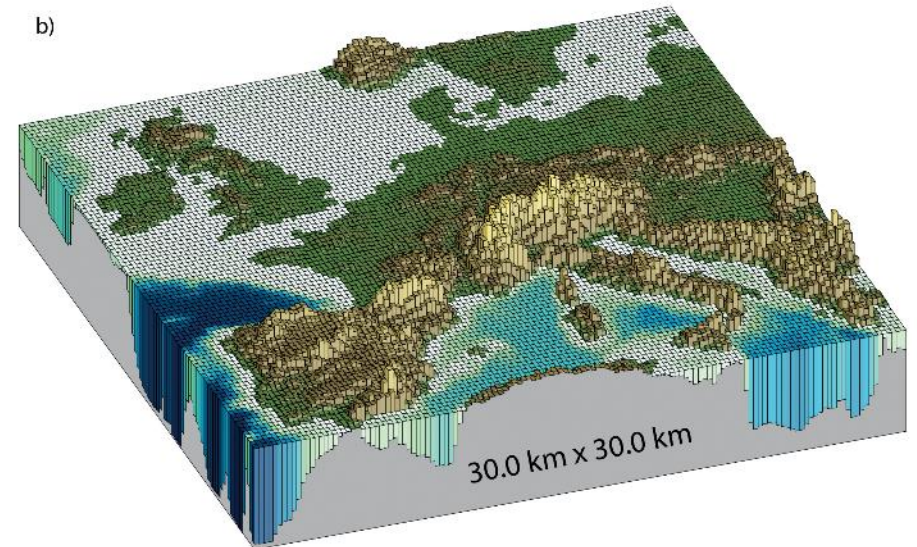
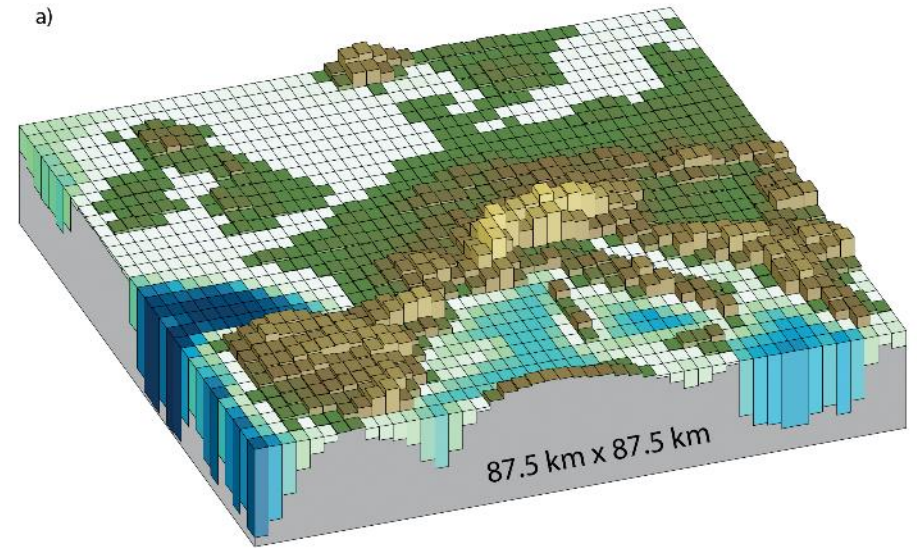
Grid boxes

- Climate models grid up the world into **boxes**
- In each grid box, properties (temperature, wind, rain, cloud cover, etc) are assumed to be uniform (mostly)



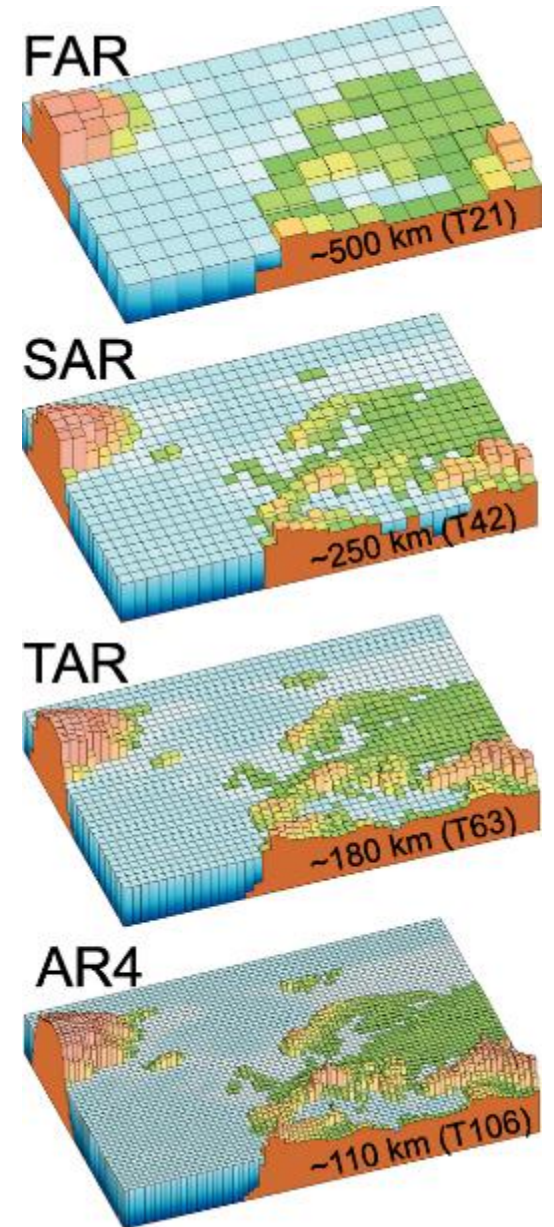
Climate model resolution

- The performance of climate models in large part depends on how small the grid boxes are
- Smaller grid boxes → higher resolution
→ better representation of the processes



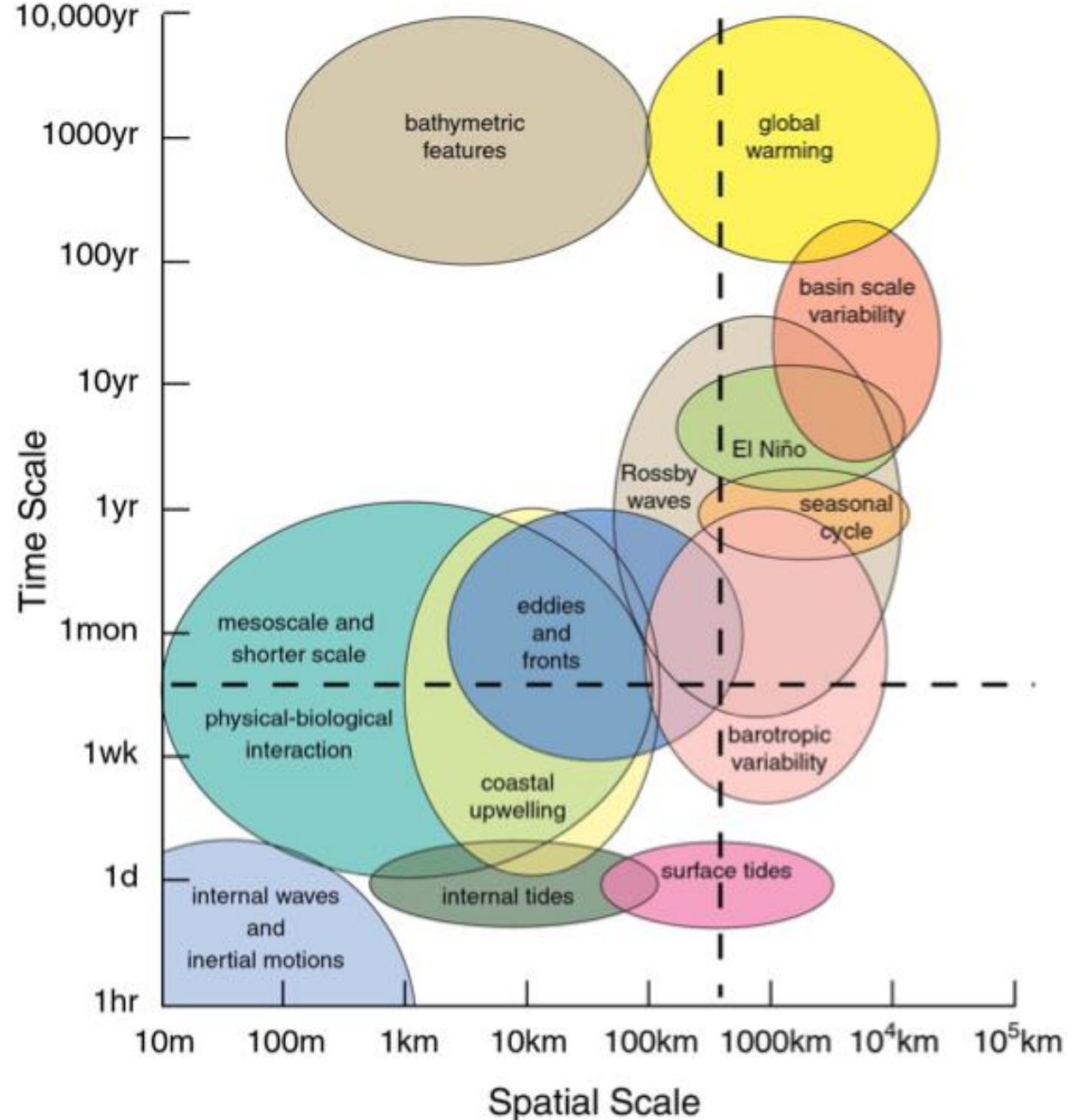
Computational cost

- Increasing complexity and resolution of climate models means constantly increasing computational cost
- Global climate models are typically run on supercomputers
- Increasing computer performance allows increasing climate model resolution
- Data storage is another challenge
 - For the IPCC AR6, >20 Petabytes of model data were generated (more than 1 million wikipedias)
 - 1 PB = 1,000,000 GB



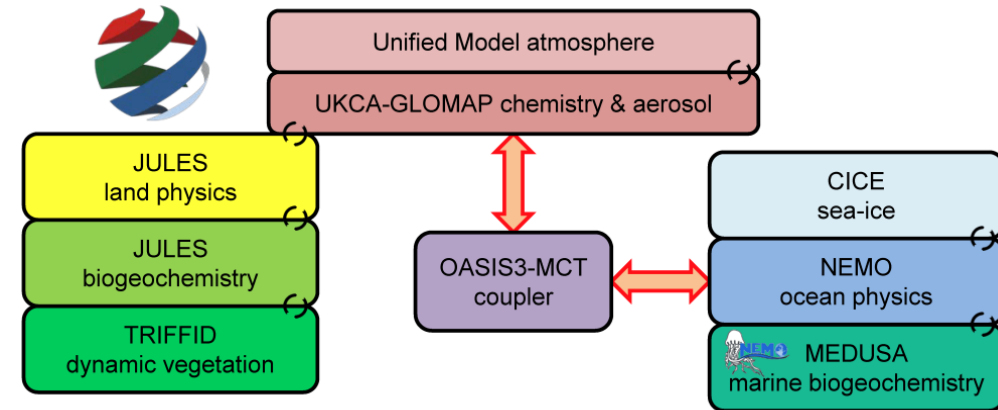
Space and time scales

- Different physical processes occur on different spatial and temporal scales
- Global climate models cannot represent all scales, particularly those smaller than the grid box size and time step duration
- Typical values for the atmospheric component in IPCC AR6 models:
 - Grid box size ~100 km across
 - Time step ~30 minutes



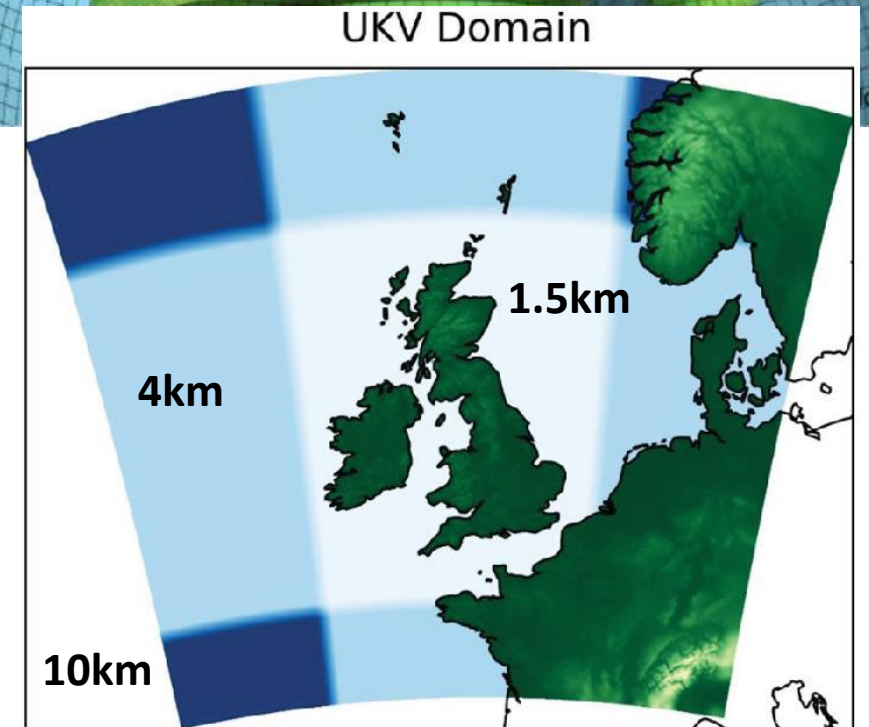
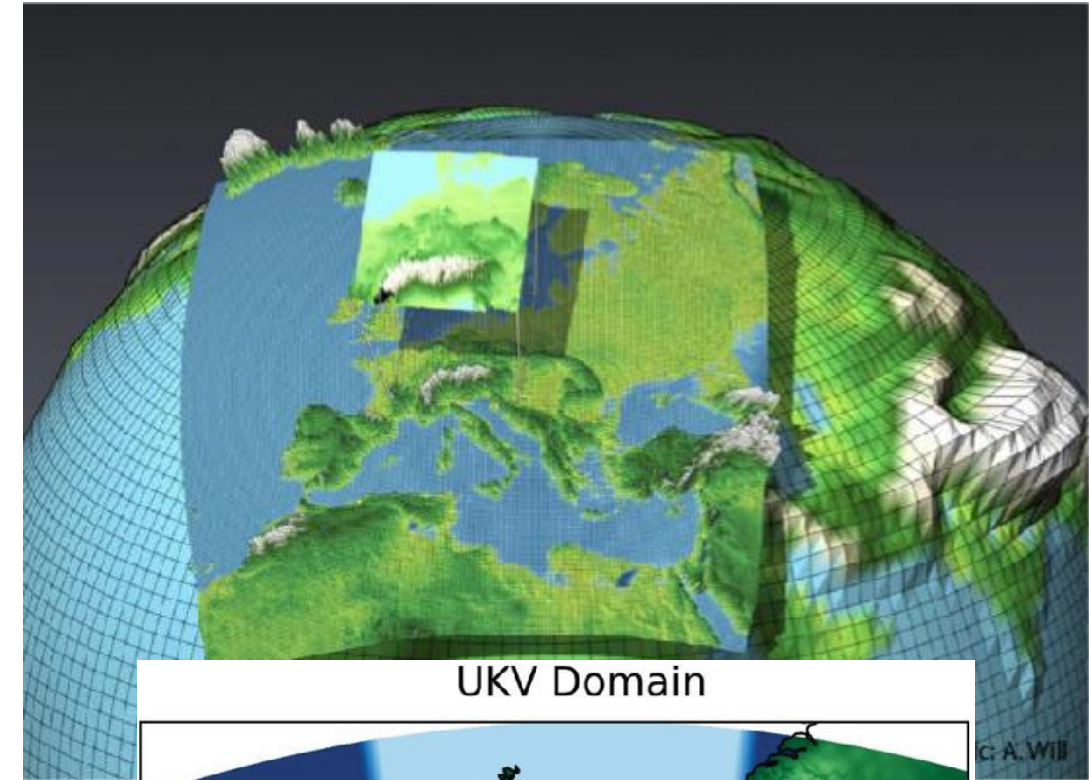
Uncoupling the model

- Often, the individual components of the climate model are run separately
- In that case, the other components are used as boundary conditions
- Example: running the atmospheric model only, with prescribed sea surface temperature as a lower boundary condition
- Advantages of running a single component:
 - Allows increased resolution
 - Easier to control resemblance to reality (see next lectures)
 - Useful for physical understanding of the processes



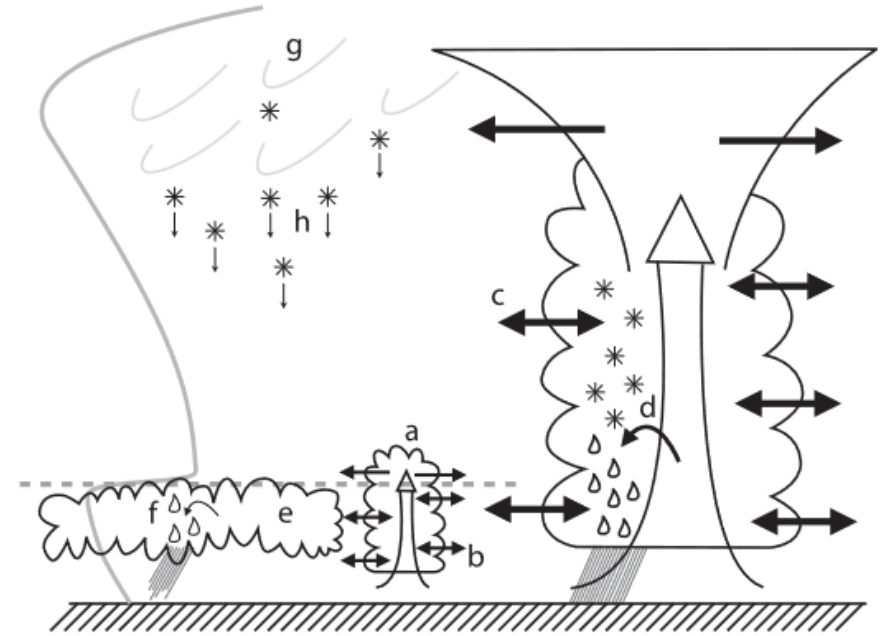
Regional climate models

- Climate models can also be run over a subset of the globe only
- This is called regional climate modelling
 - Similar to weather forecast setup
- Requires boundary conditions on the “side walls”
- Typically taken from global climate model runs
- Advantages of regional climate models are again increased resolution and ability to control forcing



Physical parameterisations

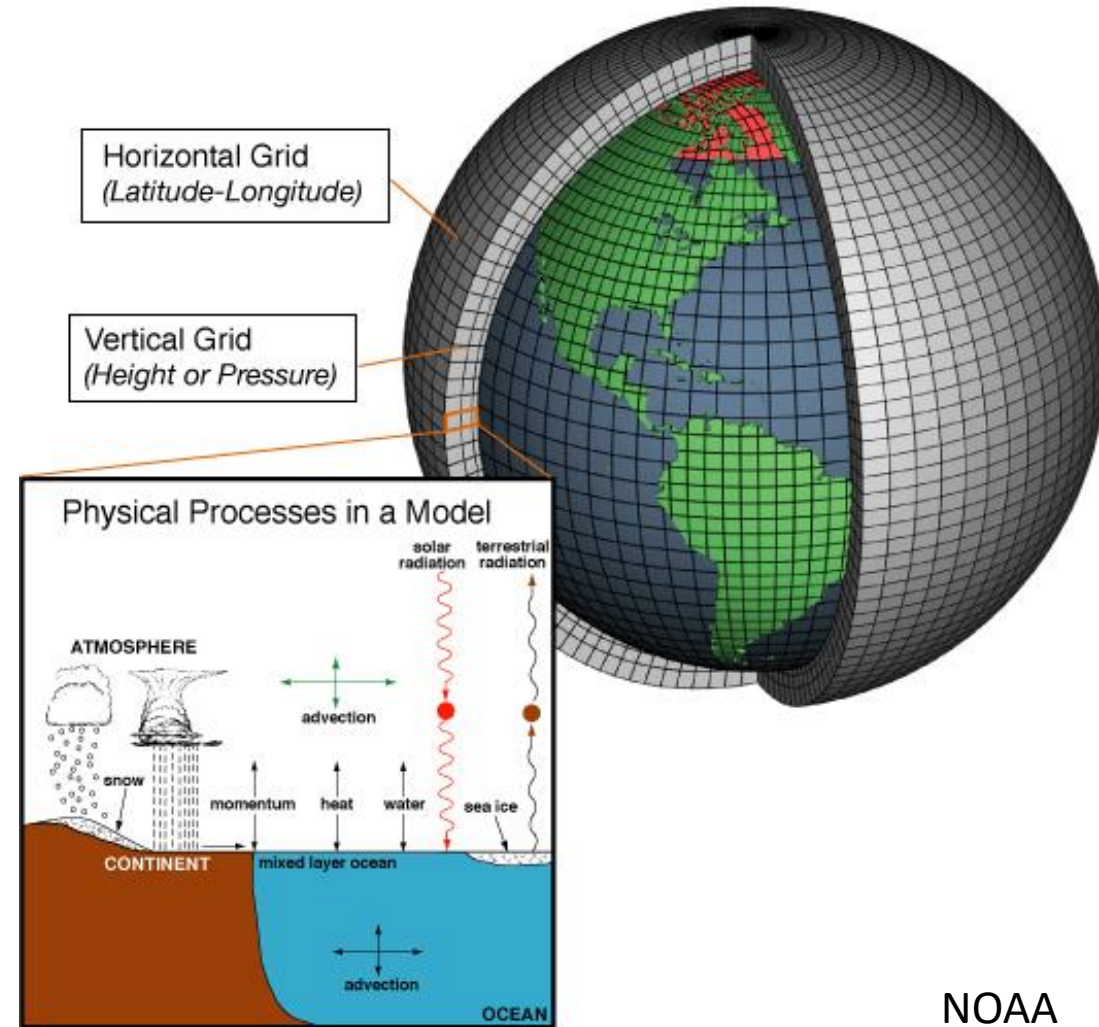
- Not all processes key to the climate can be captured by global climate models
 - Cloud processes occur on small spatial and temporal scales that can't be resolved
 - Surface drag, radiative fluxes, convection
- In this case, the climate model is limited by resolution and subgrid-scale processes need to be **parameterised**
- Parameterisations are simplifications to represent unresolved physics
 - E.g % of grid box covered by cloud - high relative humidity = high cloud cover, but what is the relationship?
- Important because parameterised processes affect global climate (see feedbacks)
- Developing good parameterisations is a major challenge in climate modelling



(some of) the processes in the ECHAM6 model

Recap: Climate modelling

- Climate models are the workhorse of climate science. They allow us to test hypotheses, and to simulate past and future climates
- Climate models solve sets of equations by stepping forward in time
- Climate models can also be run by element (e.g. only atmosphere, only ice) and in regional mode
- Climate models must parameterise important small-scale processes, e.g. clouds, that have large impacts on global climate



NOAA

Q: Why might climate models have errors?

Evaluating climate models

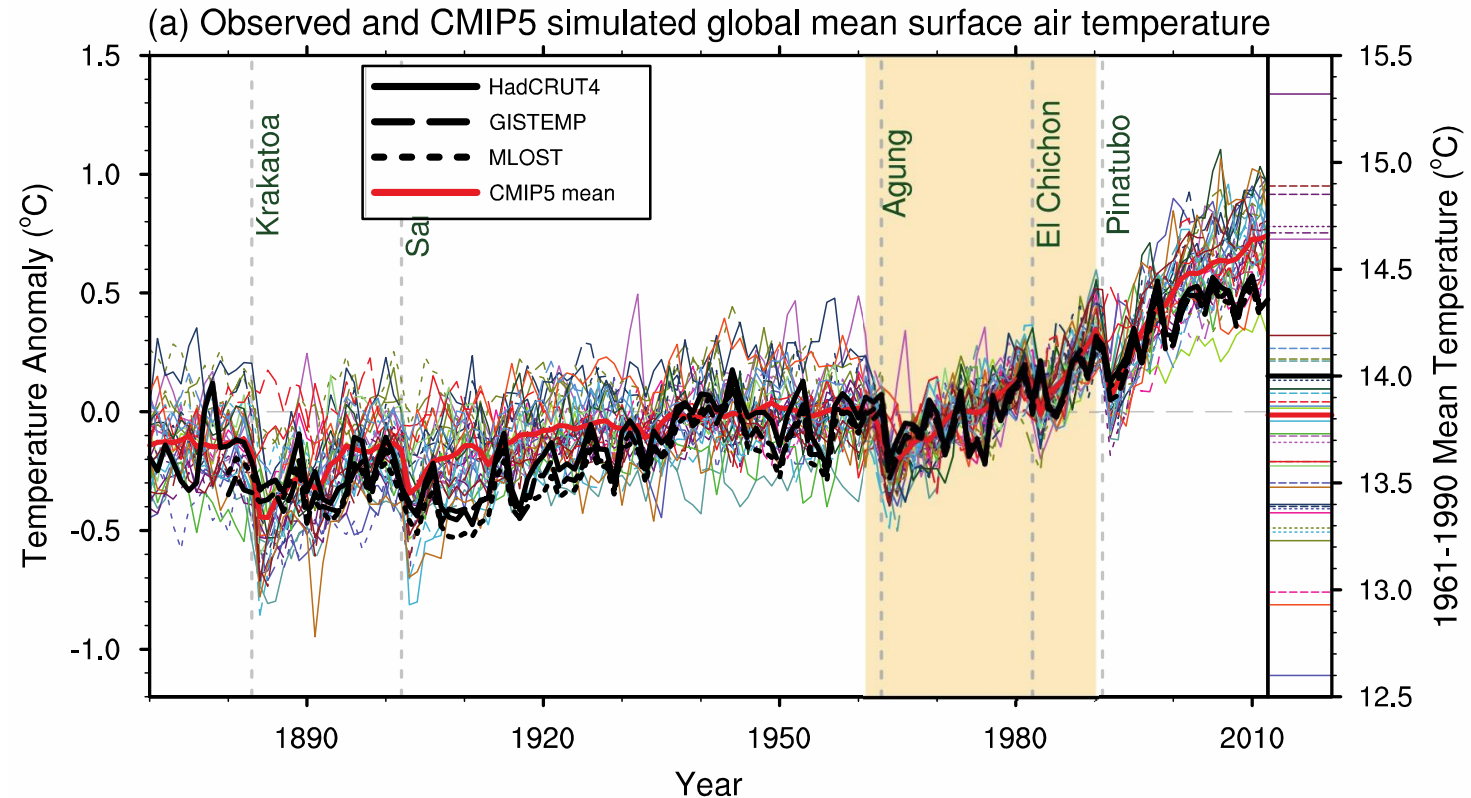
What do we look for when evaluating climate models?

- **Mean climate:** models should have the right climatology
- **Trends:** models should correctly reproduce recent climate trends (e.g., 20th century temperature)
- **Natural variability:** models should be able to simulate natural modes of variability
- **Spatial patterns:** models should get not only the global quantities right, but also their geographical patterns

These criteria are assessed by comparing models with observations

Global-mean surface temperature

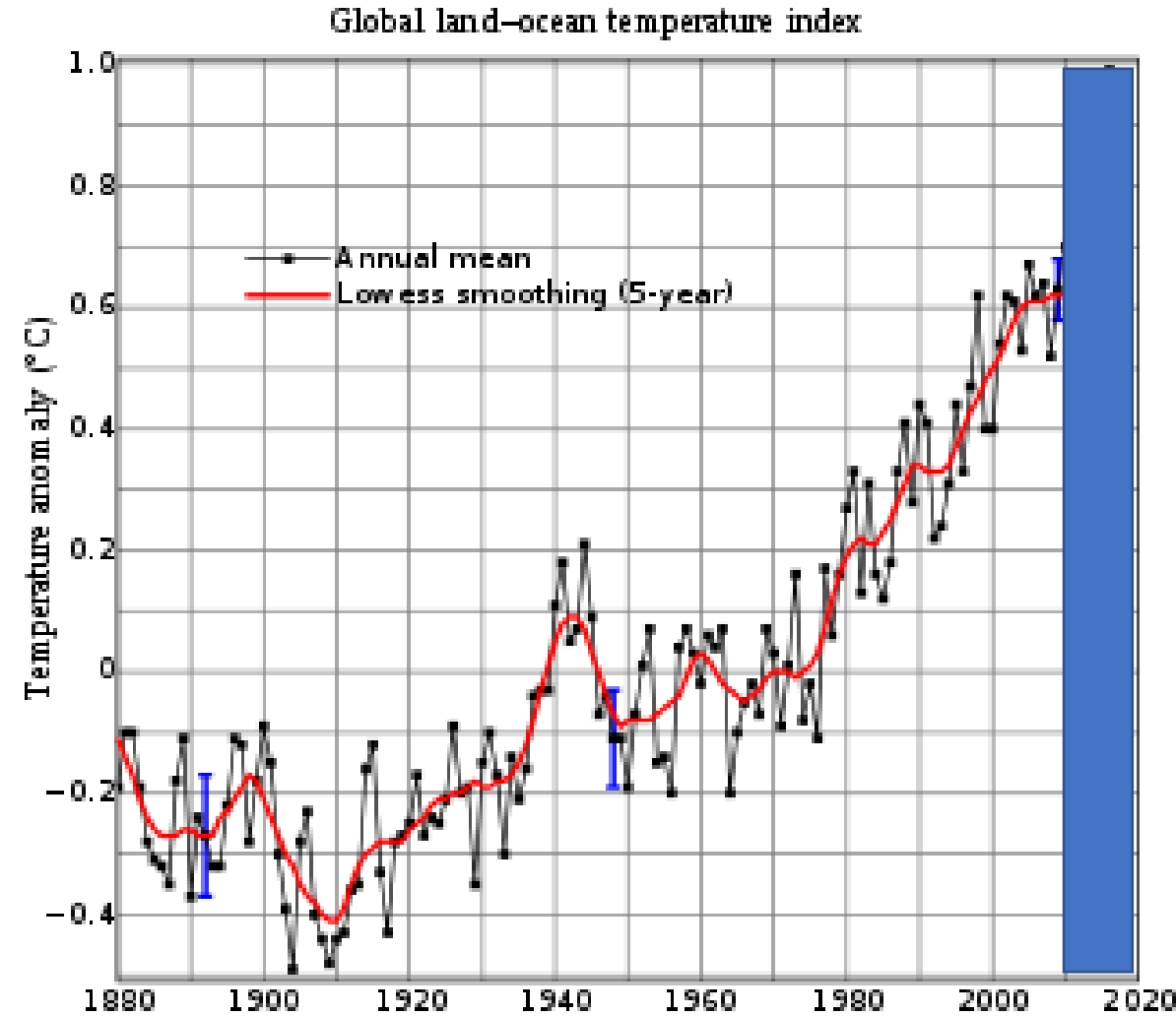
- Models simulate the evolution of global temperature reasonably well
- In the early 2000s, the observations seem to indicate a slowdown of warming (the “**global warming hiatus**”)
- This sparked debate when the AR5 came out
- However, the increase has resumed since



IPCC AR5 WG1 Chapter 9

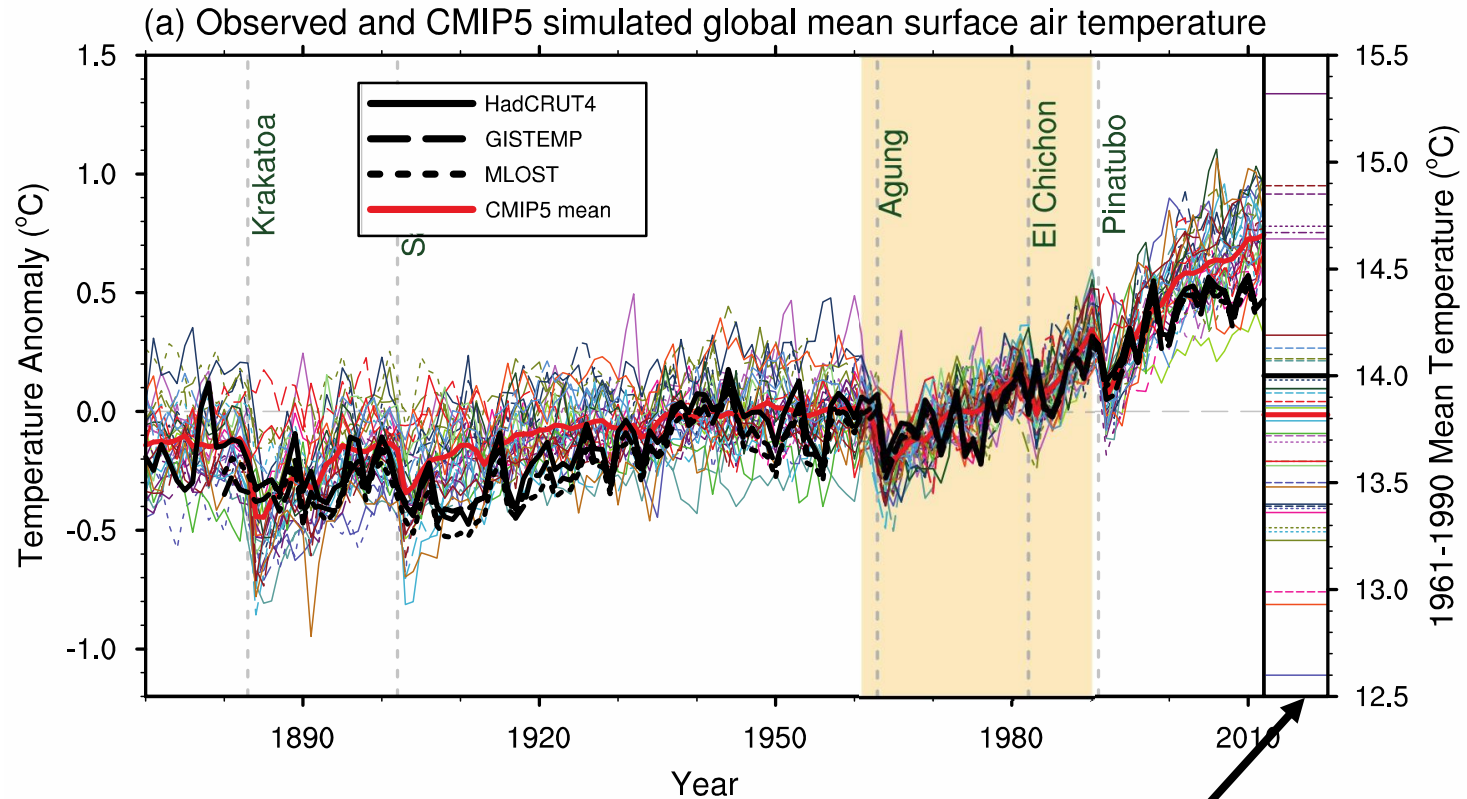
Global-mean surface temperature

- Updated global temperature record
- Rapid increase in global temperature since 2010 in observations
- The “hiatus” has ended
- Most likely it was natural variability



Global-mean surface temperature

- Model errors relative to observations are called **biases**
- Model biases of up to $\pm 1.5^\circ\text{C}$ in global-mean temperature during 1961-1990
- The mean bias is small: compensating errors among models

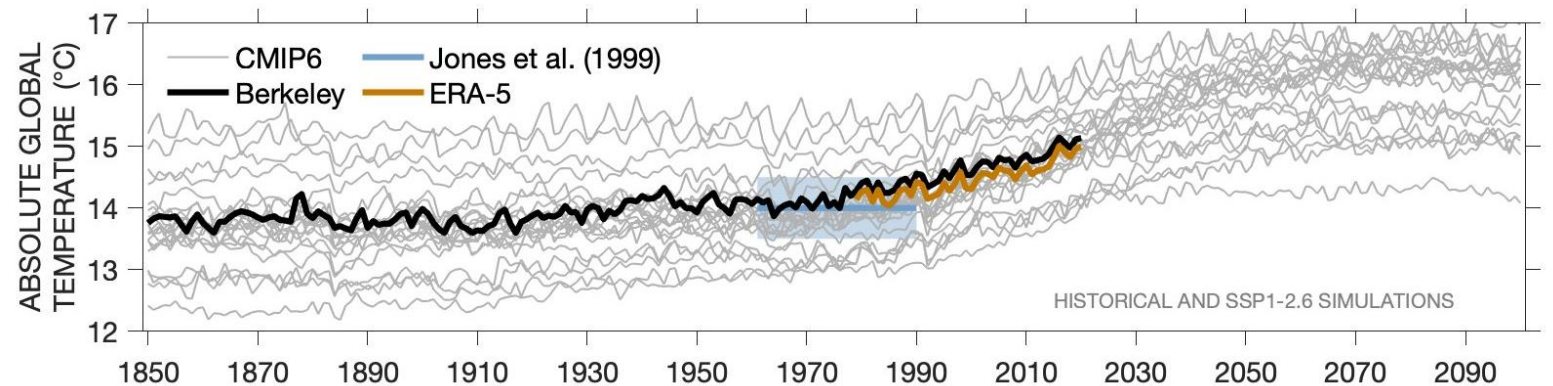
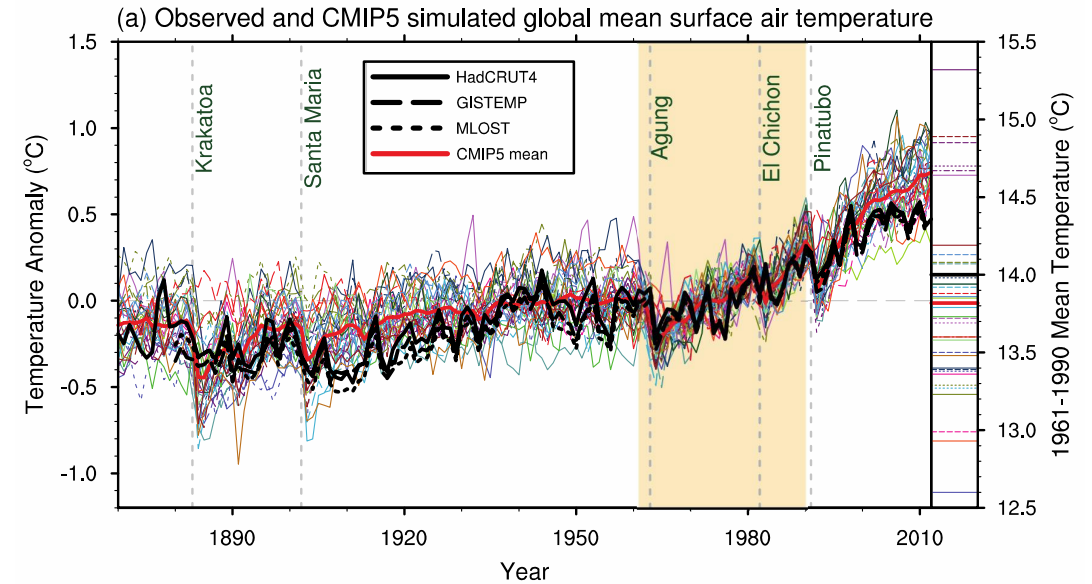


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BUT they have errors in mean temperature

Global-mean surface temperature

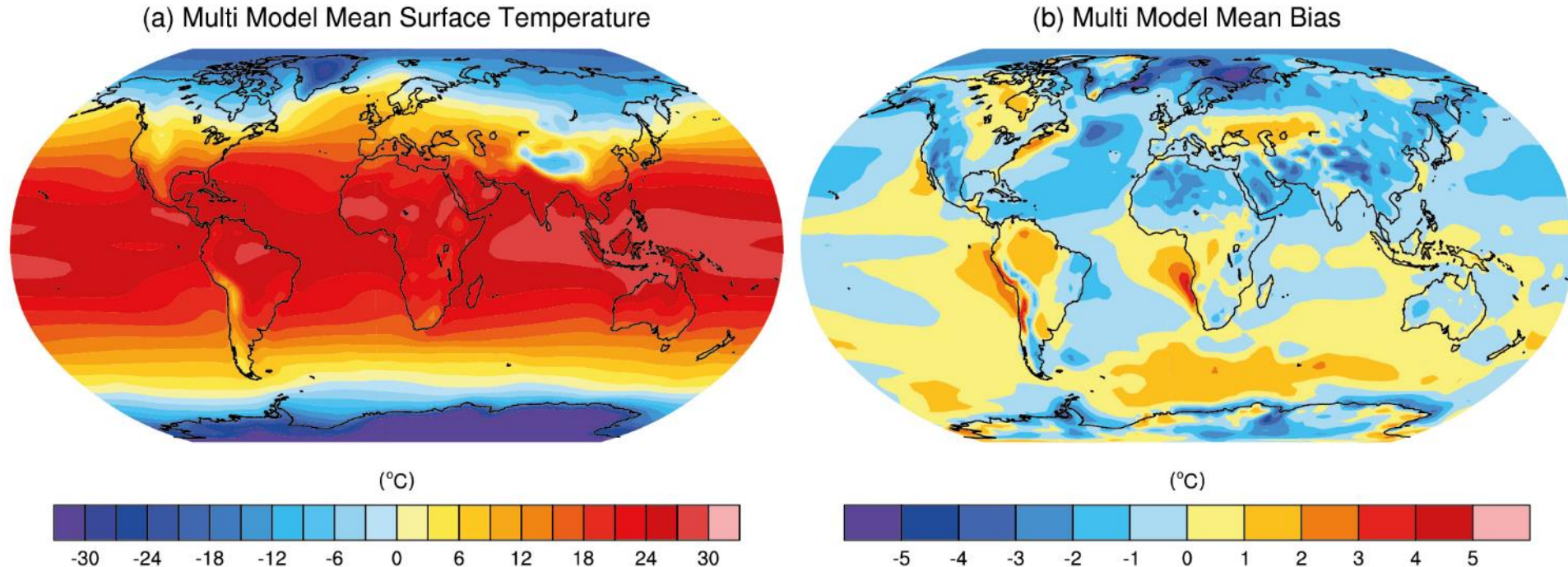
- Biases in mean temperature remain in CMIP6...
- **Q: Is this important?**
 - What processes depend on absolute temperature?
 - Are there non-linearities?



IPCC AR6 WG1 Fig. 1.11

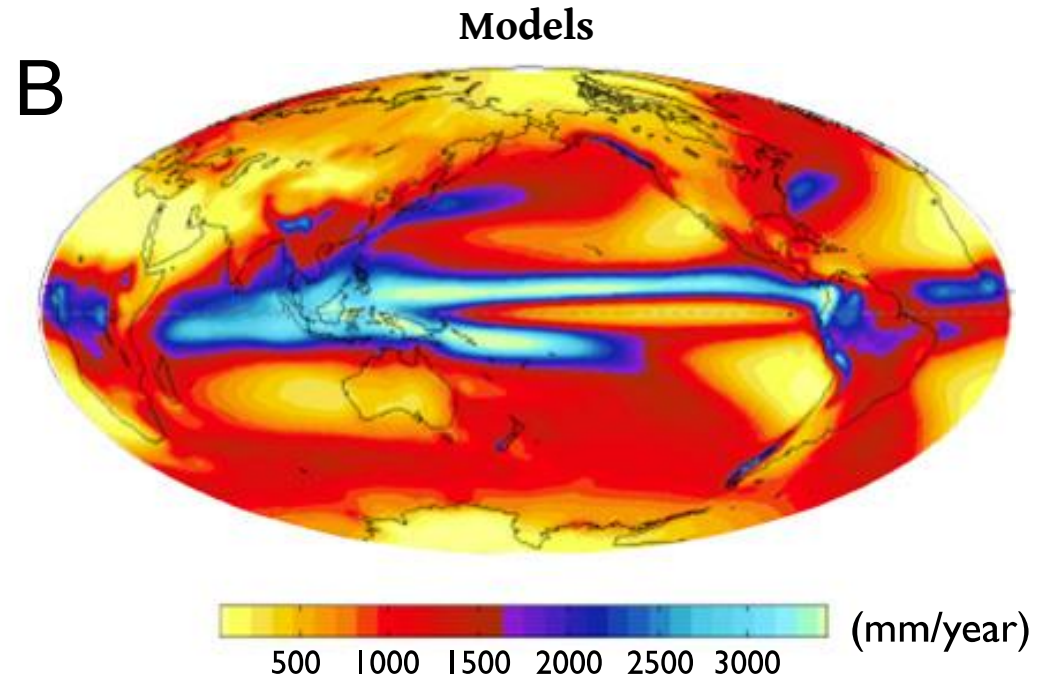
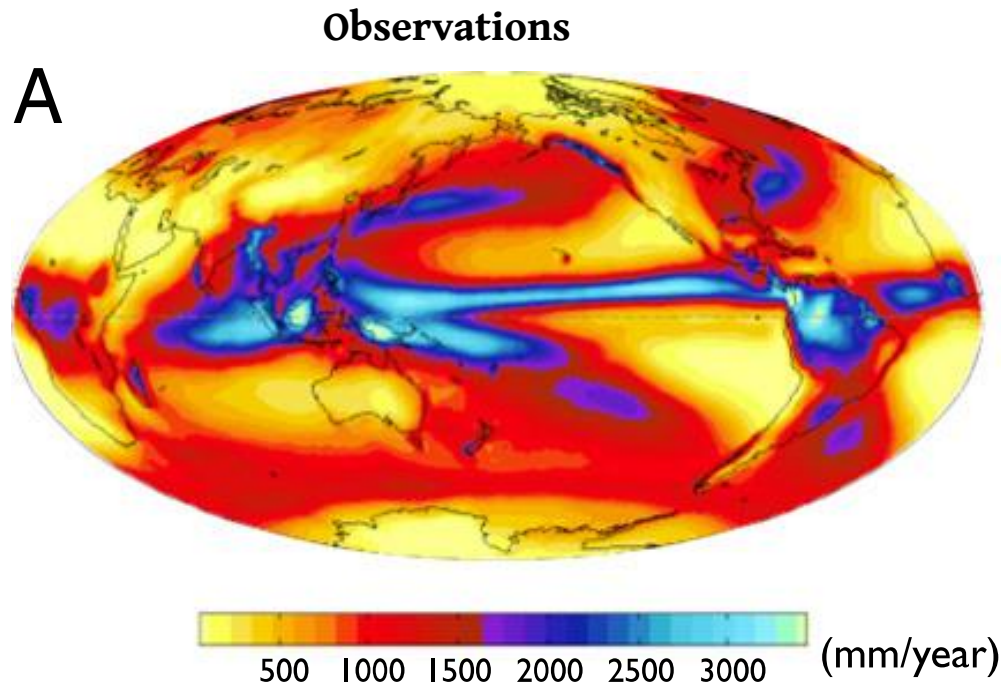
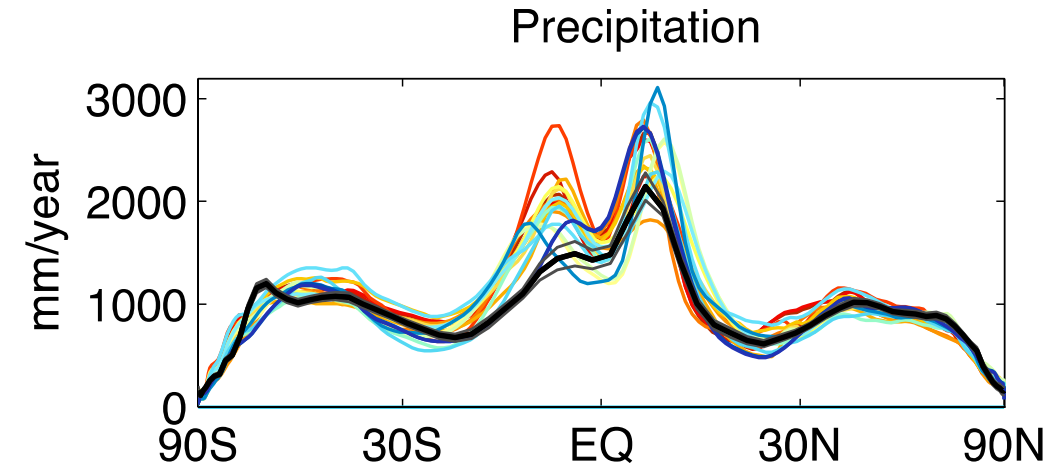
Surface temperature pattern

- Regionally, mean biases of several degrees Celsius
- Especially at eastern edge of subtropical oceans. This is linked to clouds
- Errors in ocean circulation likely important also



Precipitation

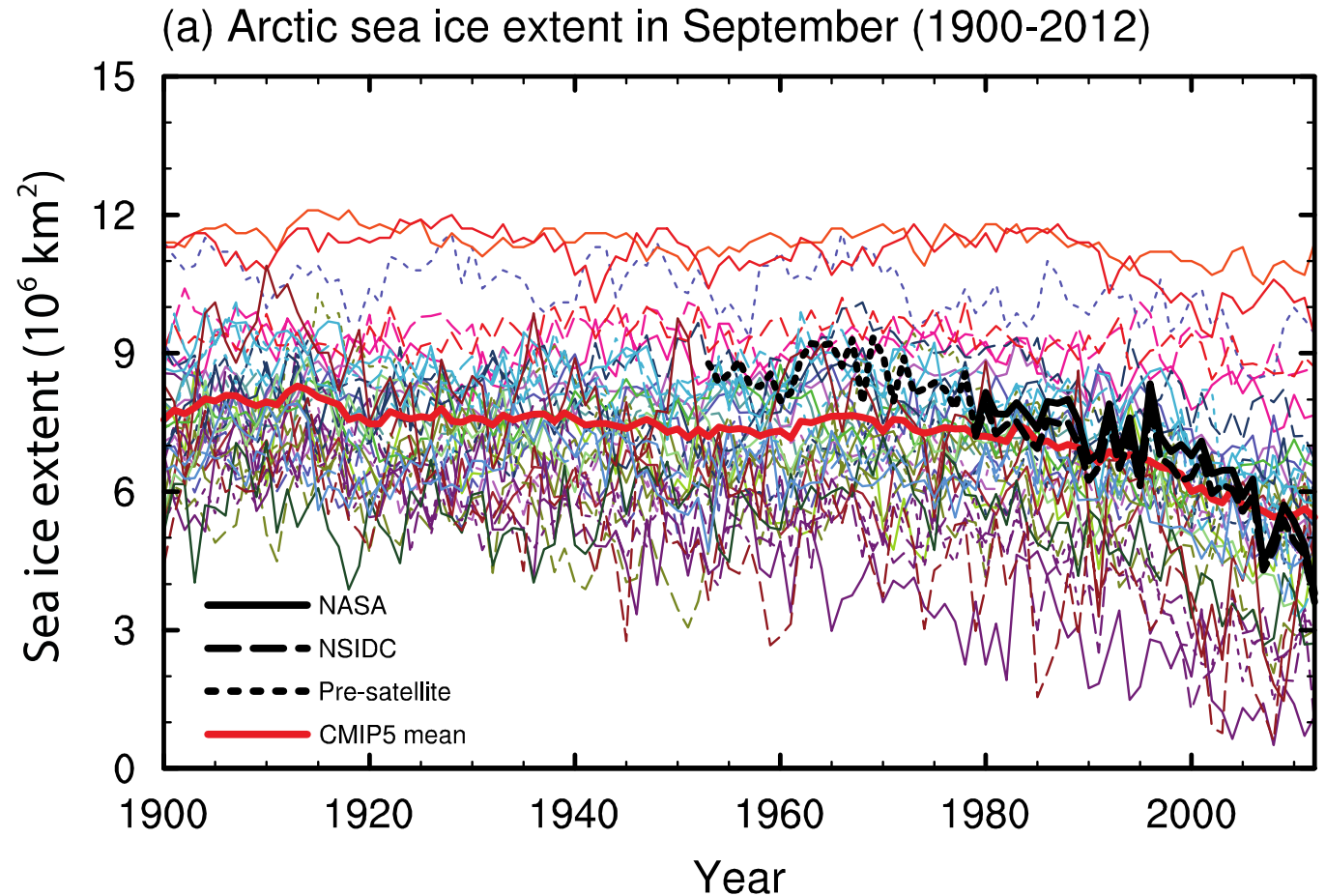
- Climate models also have biases in tropical rainfall
- Models have too much rainfall south of Equator



Hwang and Frierson 2013

Arctic sea ice

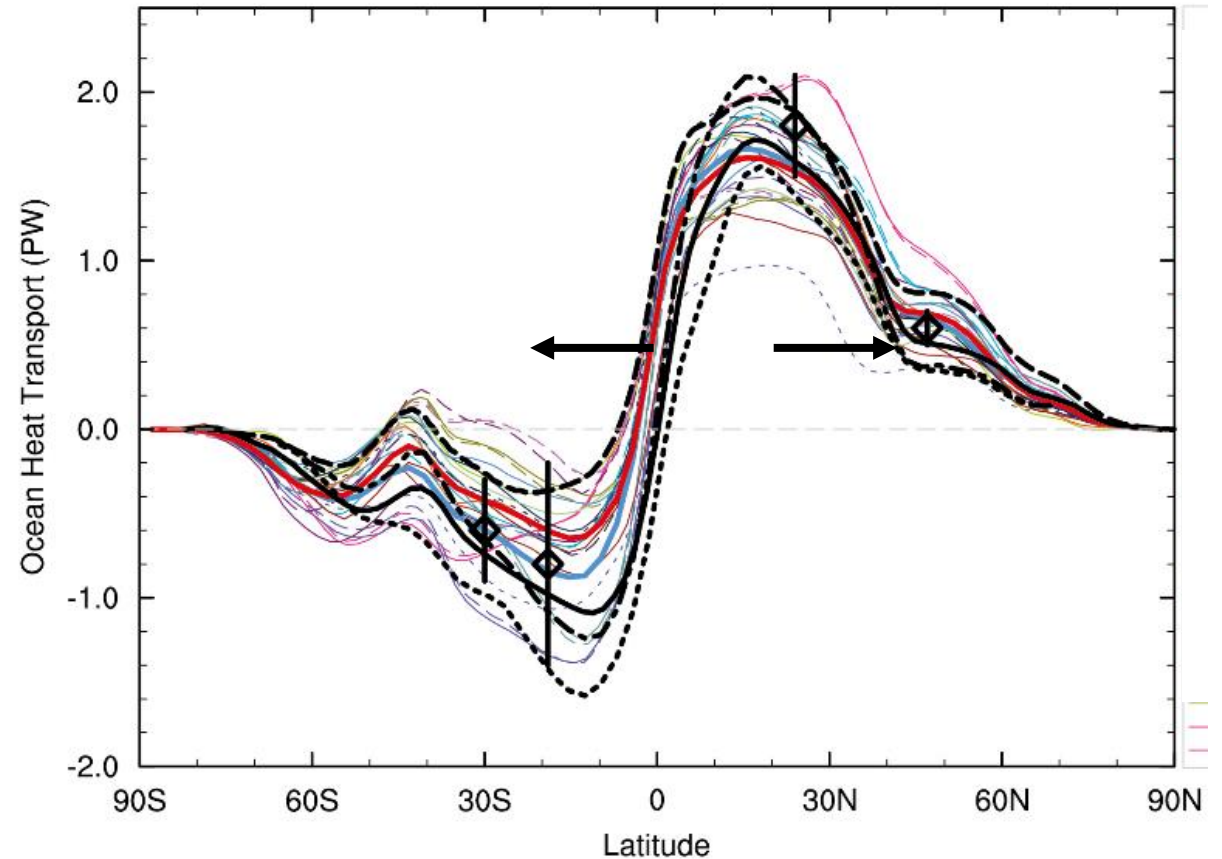
- Plot shows September sea ice extent (the annual minimum)
- Large biases in models: some have far too much sea ice, others are almost ice-free already
- Again the multi-model mean looks reasonable



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Ocean heat transport

- Models also have errors in ocean circulation and therefore ocean heat transport
- Here the observations are also quite uncertain, so it's hard to tell which models are closer to reality



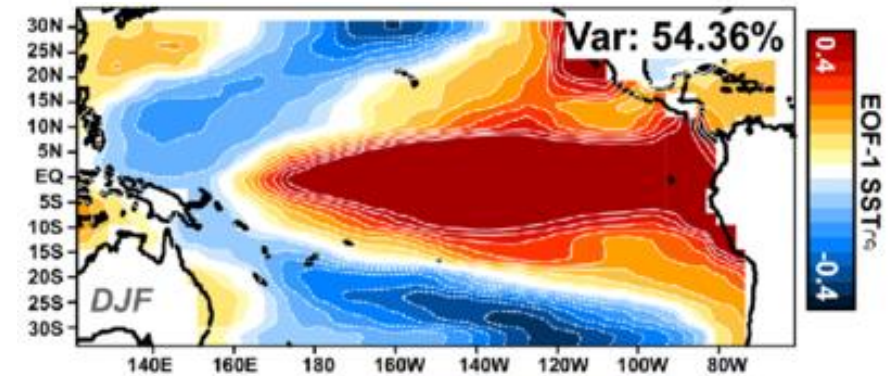
Black: observational estimates
Red: CMIP5 mean

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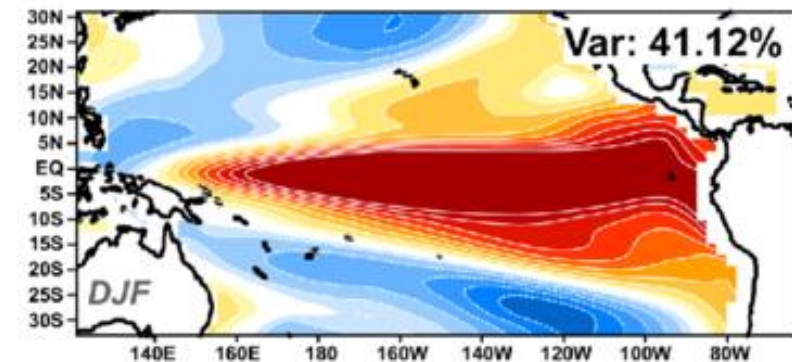
ENSO

- Models are evaluated not just based on the mean climate and on trends, but also based on natural variability
- A good model should have realistic variability
- Climate models are generally able to simulate ENSO, but there are some biases in sea surface temperature
- This also means biases in local and remote ENSO impacts (temperature, rainfall)

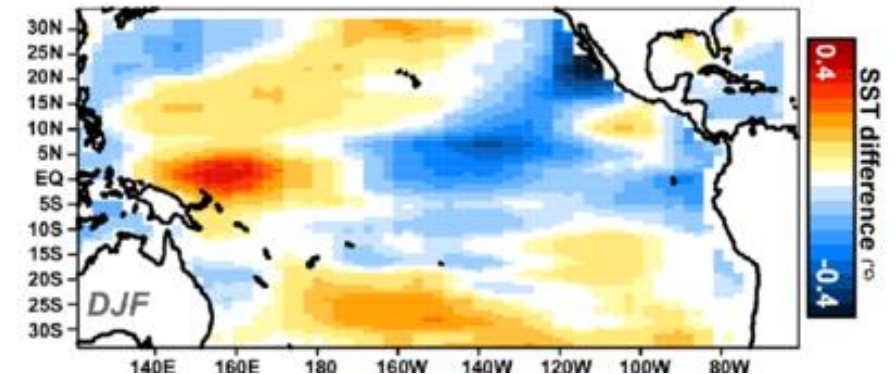
Observations



CMIP5 models



Models minus observations

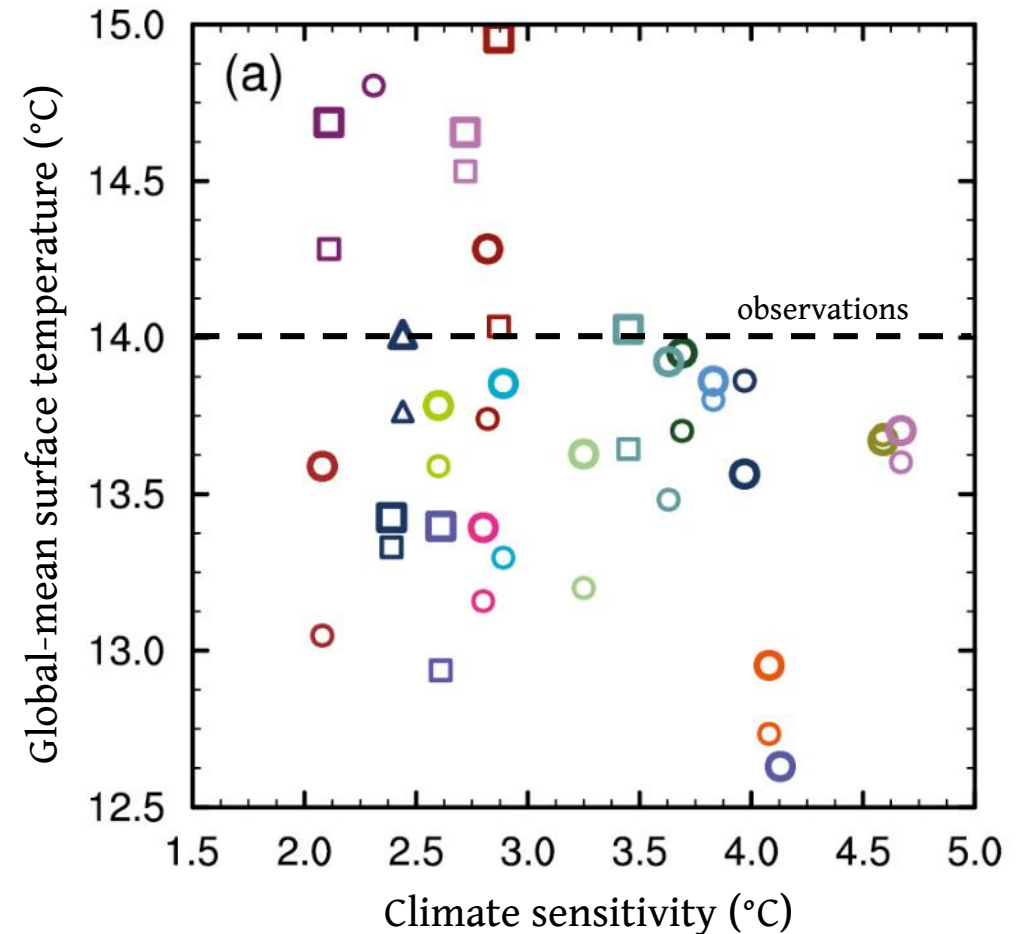


The role of biases for future projections

- The underlying rationale of validating models against observations is
 - *“The smaller the model bias, the better the model”*
- But, unclear whether smaller bias translates into more accurate climate projections!
 - Q: Why might this be?
- This is one of the reasons why IPCC usually publishes simple, unweighted multi-model-means

The role of biases for future projections

- Models predict a wide range of climate sensitivity values. Is this because of mean-state biases?
- There's no obvious relationship between historical temperature and climate sensitivity in models
- So it's unlikely that reducing biases in global temperature would reduce the range of climate sensitivity in models

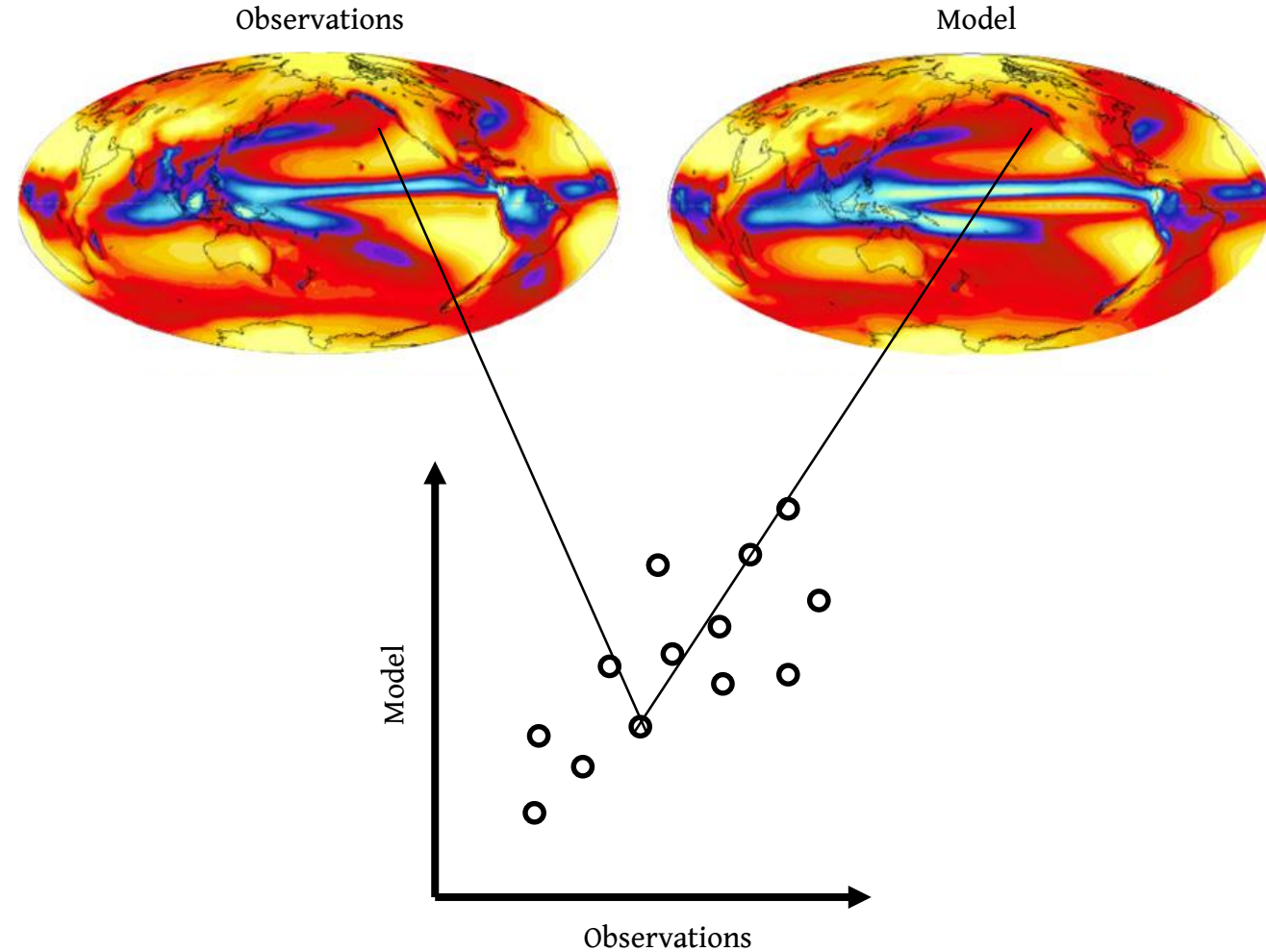


Objective measures of climate model performance

- It is useful to assess the performance of climate models objectively (i.e., statistically)
- This allows us to compare models with one another, or to assess how changes in the model code affect the model performance
- Some commonly used metrics:
 - Pattern correlation
 - Root mean-square error (RMSE)

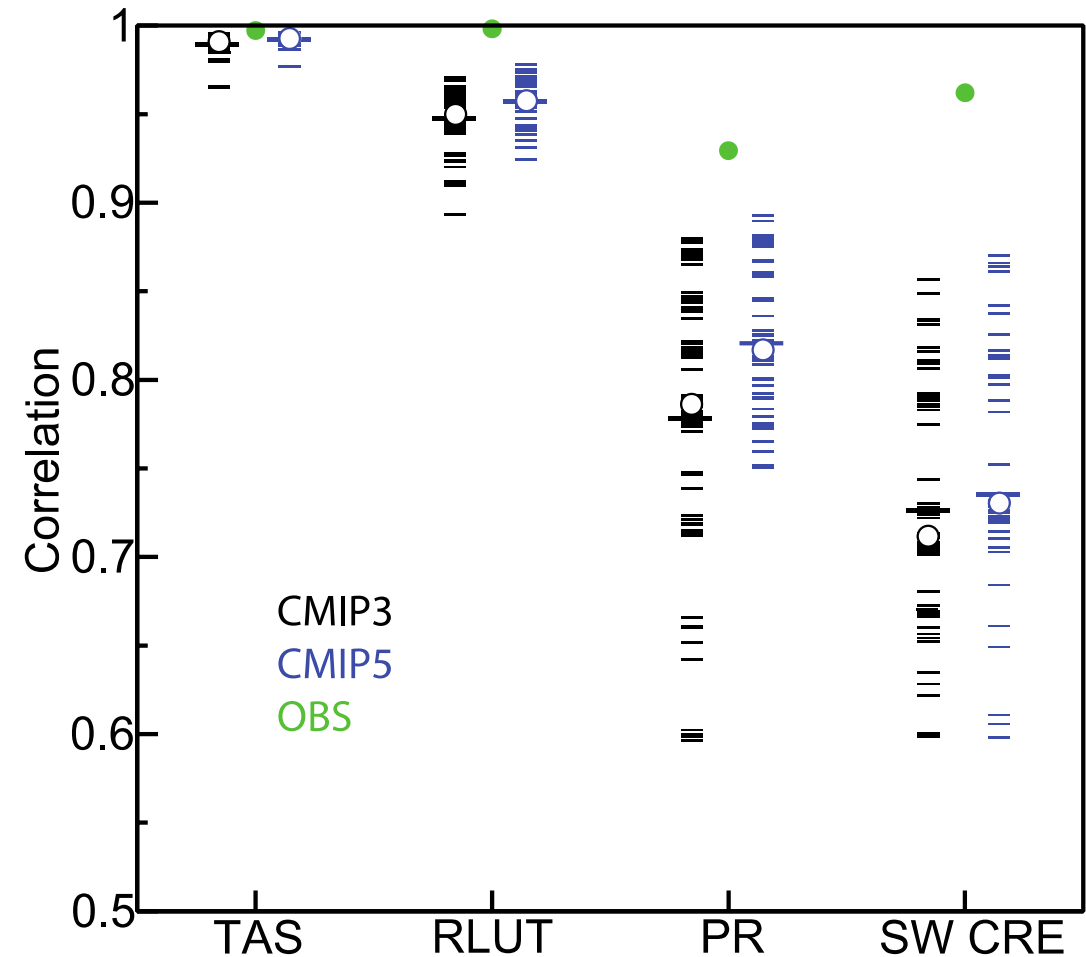
Pattern correlation

- The correlation between two patterns
- Typically you would compare models with some reference observation dataset
- High correlation means that the model can reproduce the observed pattern well
- Can do this for every model and compare the values



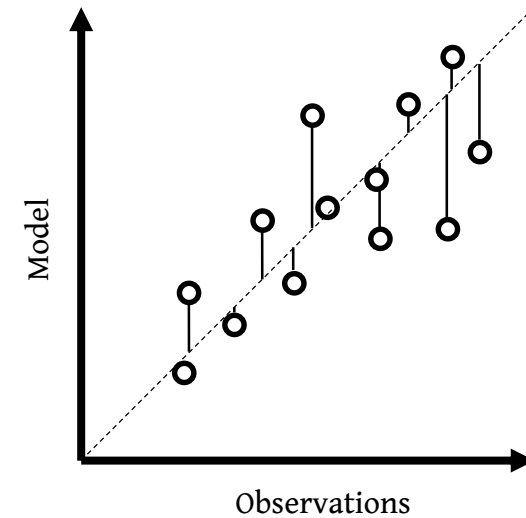
Pattern correlations in CMIP3 and CMIP5

- Pattern correlations in CMIP5 models
- TAS: surface temperature
- RLUT: outgoing longwave radiation
- PR: precipitation
- SW CRE: shortwave cloud-radiative effect
- Generally, CMIP5 models perform better than CMIP3 models
- Green dot shows correlation between two distinct observational datasets



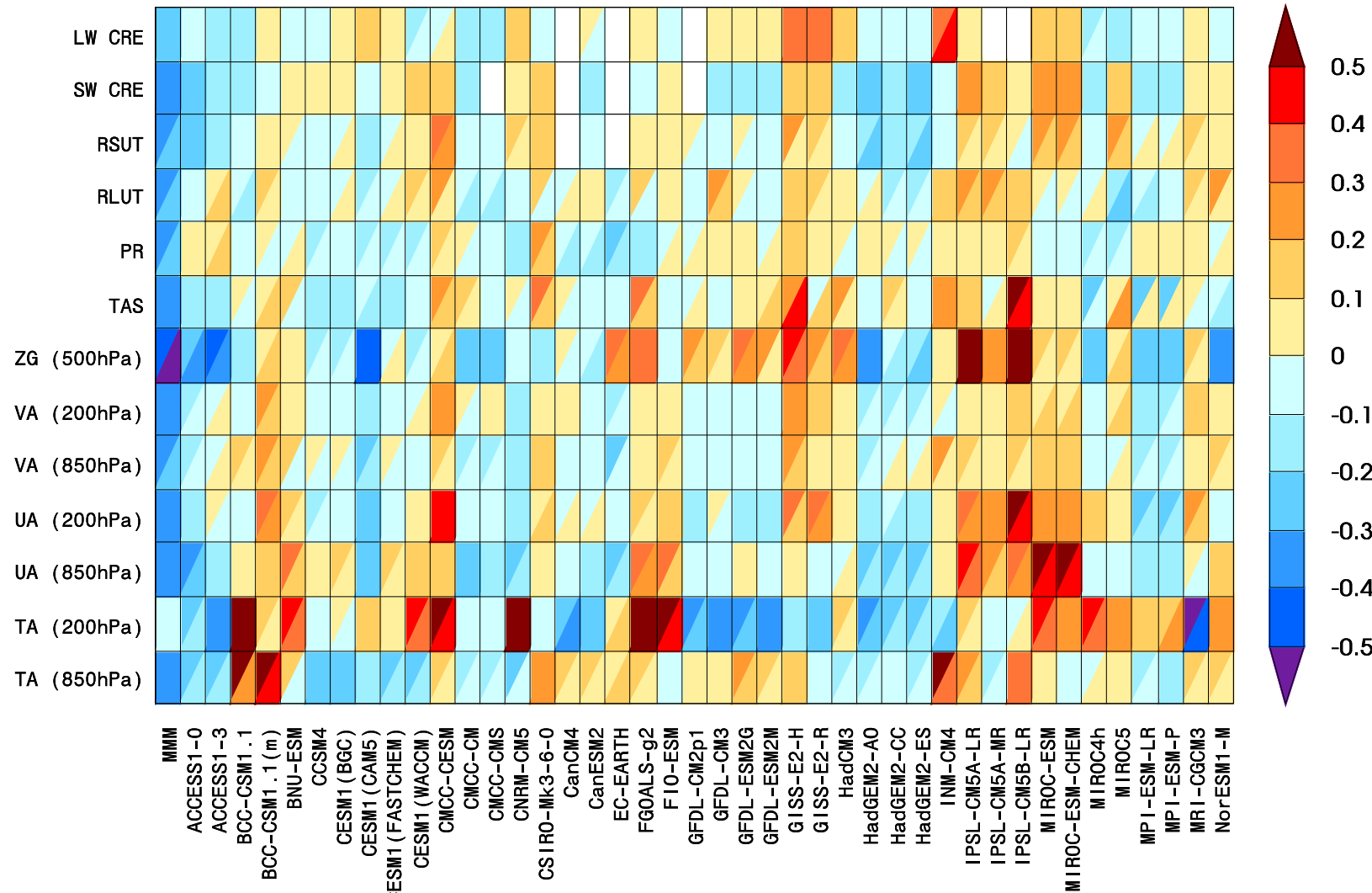
Root mean-square error (RMSE)

- A measure of the size of the vertical bars
- Larger bars = larger error = higher RMSE
- If the model were perfect, the dots would lie on the dashed line (one-to-one)



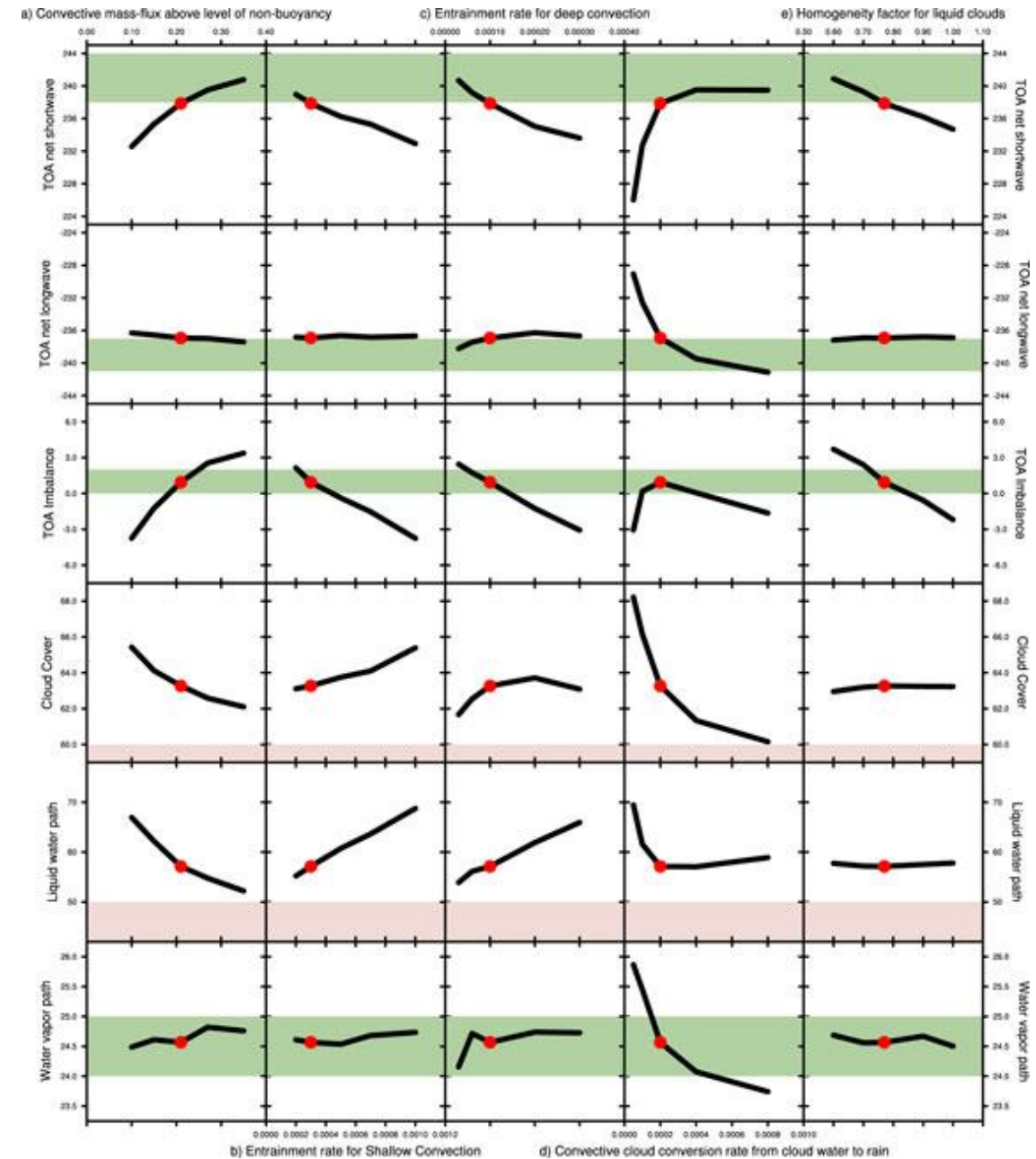
RMSE in CMIP5 models

- RMSE by variable and model
- Blue is smaller, red is larger
- Some models perform better than others
- The multi-model mean performs better than any single model



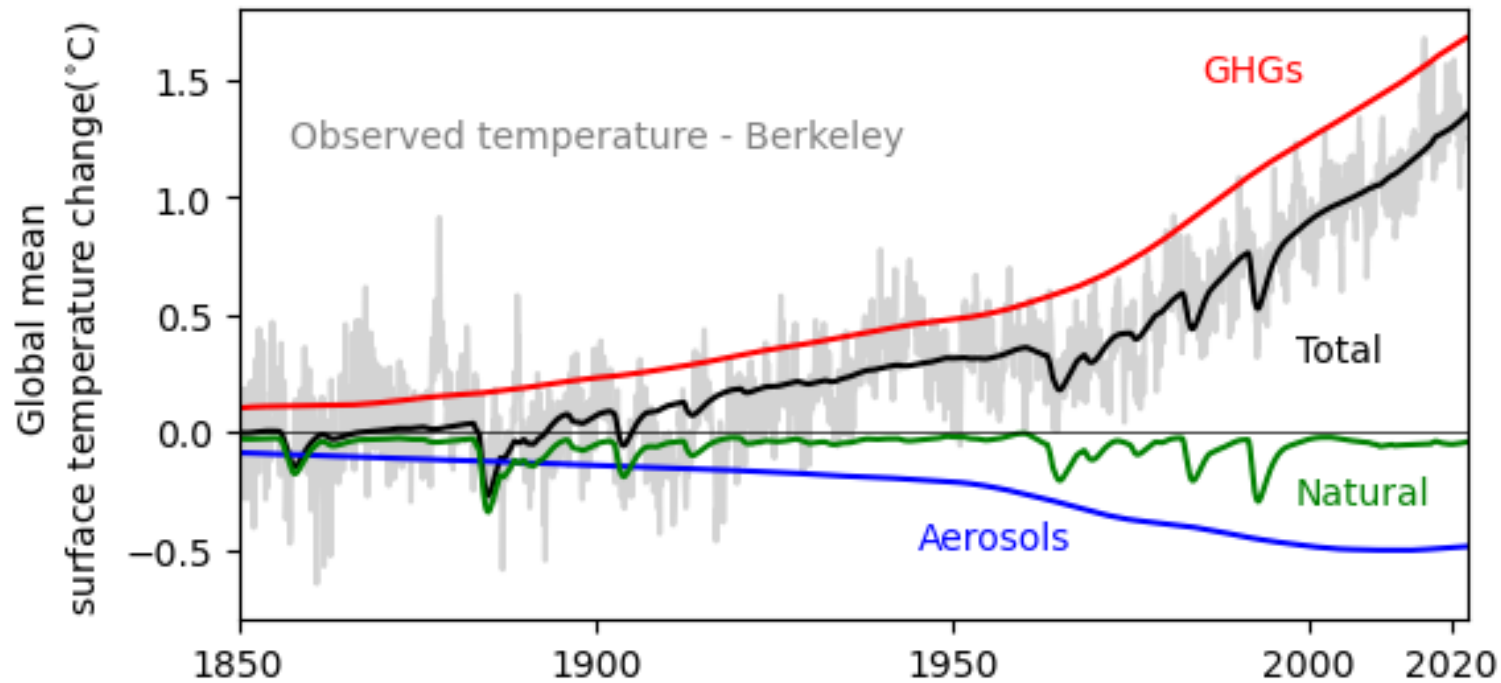
Model tuning

- Climate model parameterisations use many parameters whose values are uncertain
 - Cloud parameters
 - Surface roughness parameters
 - Ocean mixing parameters
- Because the values are uncertain, model developers optimise these parameters to reduce model biases in historical climate
- This is known as **model tuning**
- Are optimal parameters for historical climate optimal for a future warmer climate?



Model tuning

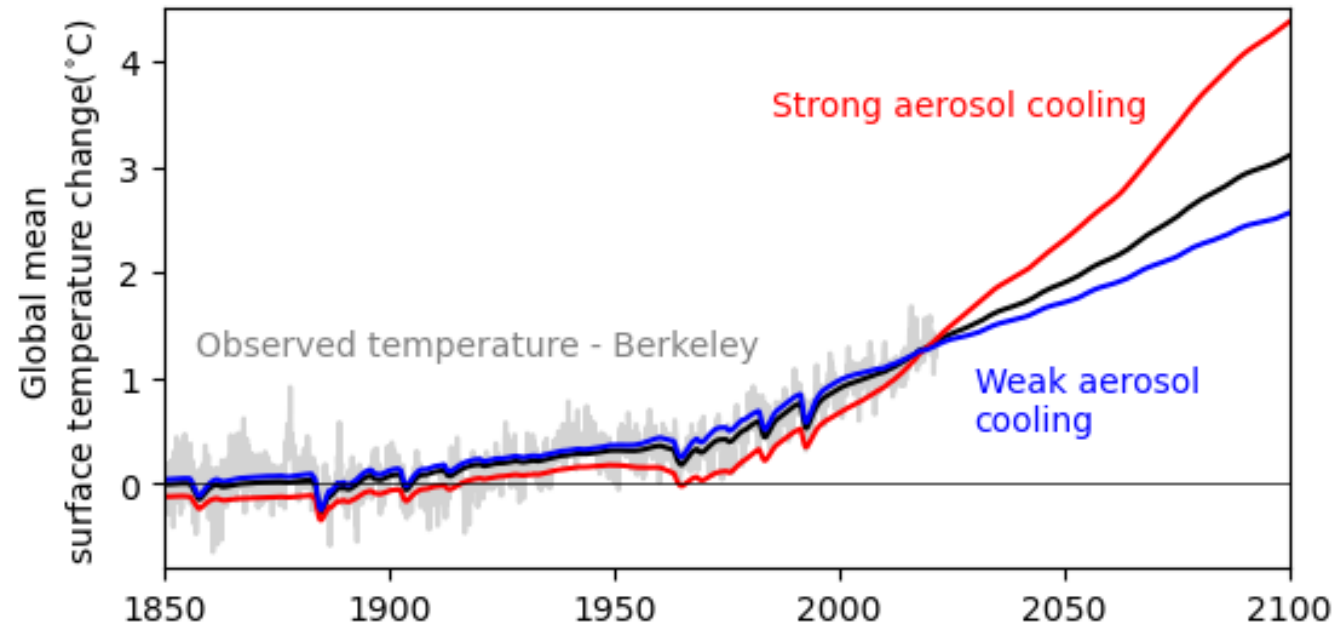
- Typical tuning targets are global climatologies of temperature, radiative fluxes, clouds
- Also spatial patterns of the same variables
- Most models are not tuned for given climate sensitivity values or historical trends, but they sometimes can be.



A simple model, tuned to reproduce the twentieth century
www.cloudsandclimate.com

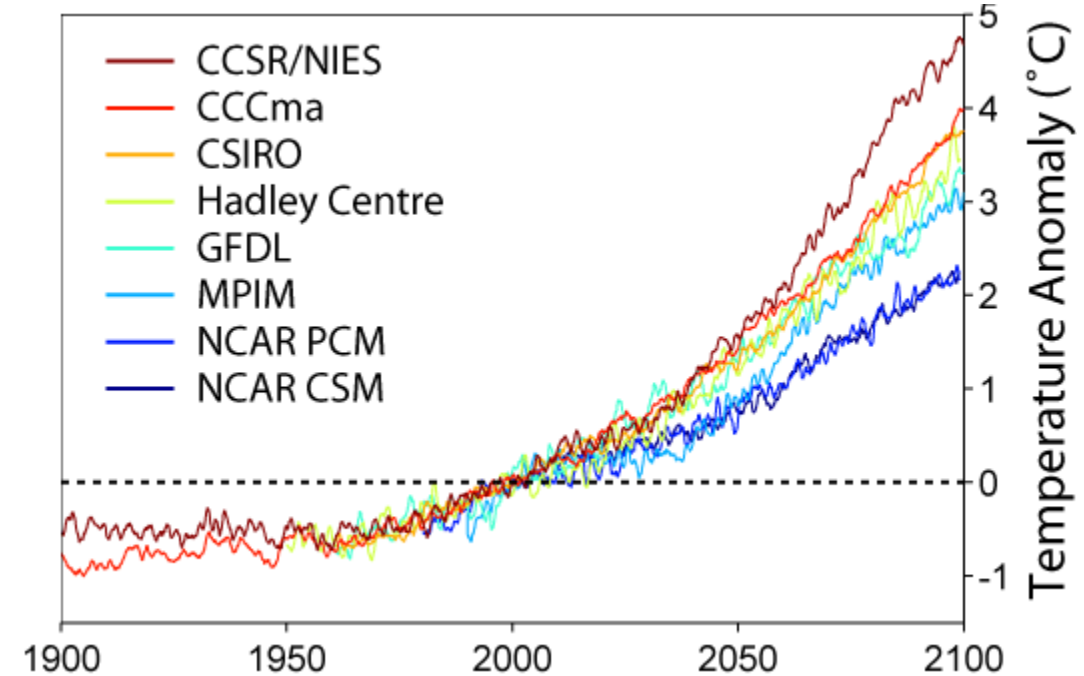
Past/future climate

- Matching past climate change does not necessarily constrain future projections
- You can reproduce 20th century temperature with either:
 - High ECS, high aerosol forcing (red)
 - Low ECS, low aerosol forcing (blue)
- We need extra constraints on model behaviour to narrow the uncertainty
 - **Q: What should we look at?**



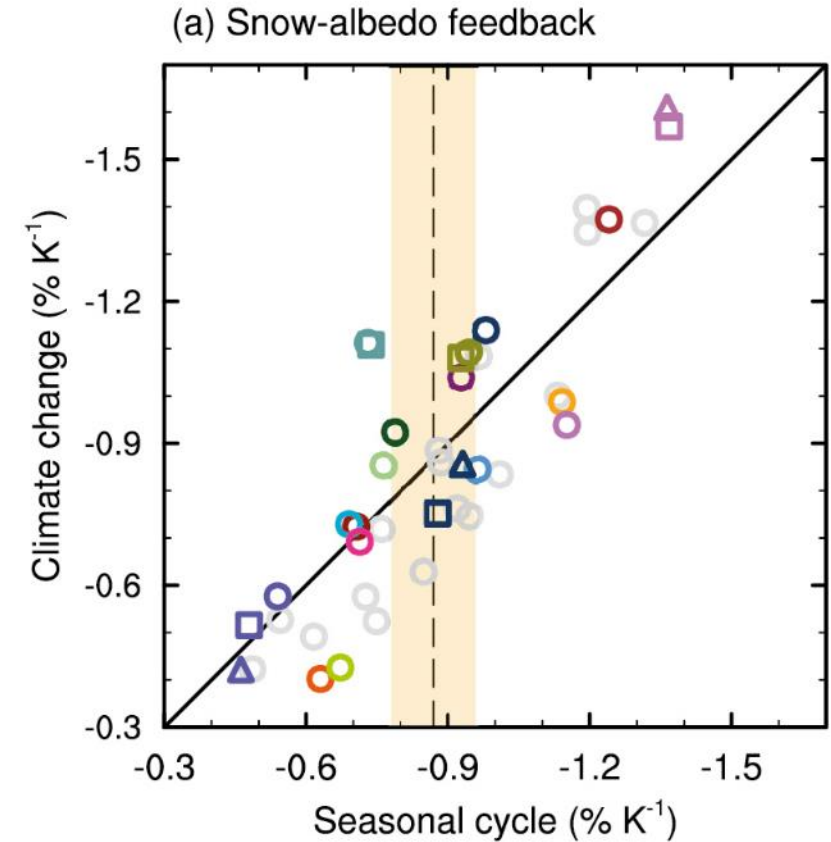
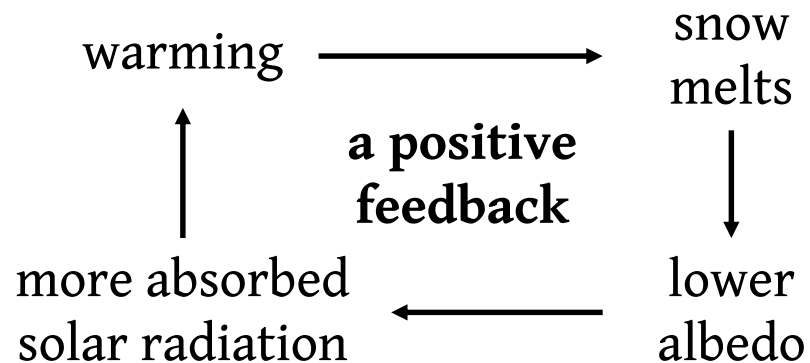
Constraining model behaviour

- Because of model error, models give us a broad range of projections for a given scenario, causing uncertainty
- Reducing model error is hard
- Possible way around this problem:
 - Find an observable relationship that tells us something about future climate change
 - Use this relationship to rule out some of the models, based on observations
- This is called an “emergent constraint”



Emergent constraints

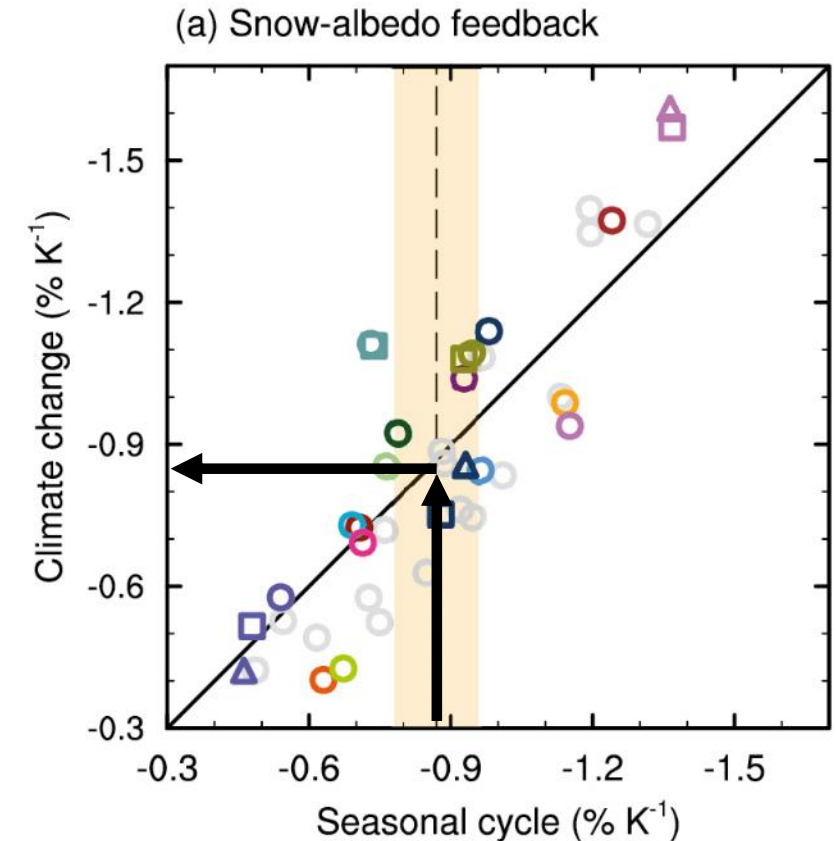
- Example: snow-albedo feedback
- Defined here as the percent albedo change per degree warming
- When temperature increases, snow cover decreases and albedo goes down
- The numbers are negative because we are looking at the albedo changes, but this is a positive *radiative* feedback:



IPCC AR5 WG1 Chapter 9

Emergent constraints

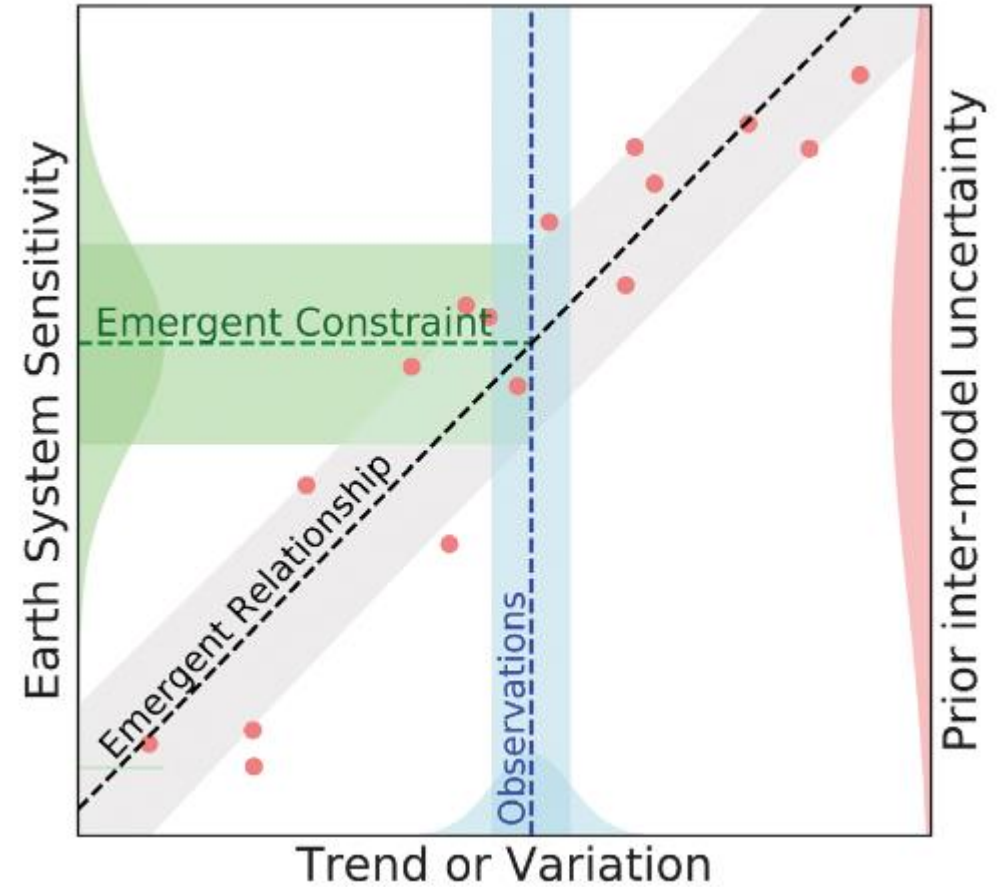
- In climate models, the seasonal feedback is a very good predictor of the climate change feedback
 - Models with a large seasonal feedback have a large climate change feedback, and vice versa
- The seasonal feedback can be estimated with good confidence (orange band)
- This allows us to rule out many of the model projections, and to provide a most likely value for the real snow-albedo feedback in future climate change



IPCC AR5 WG1 Chapter 9

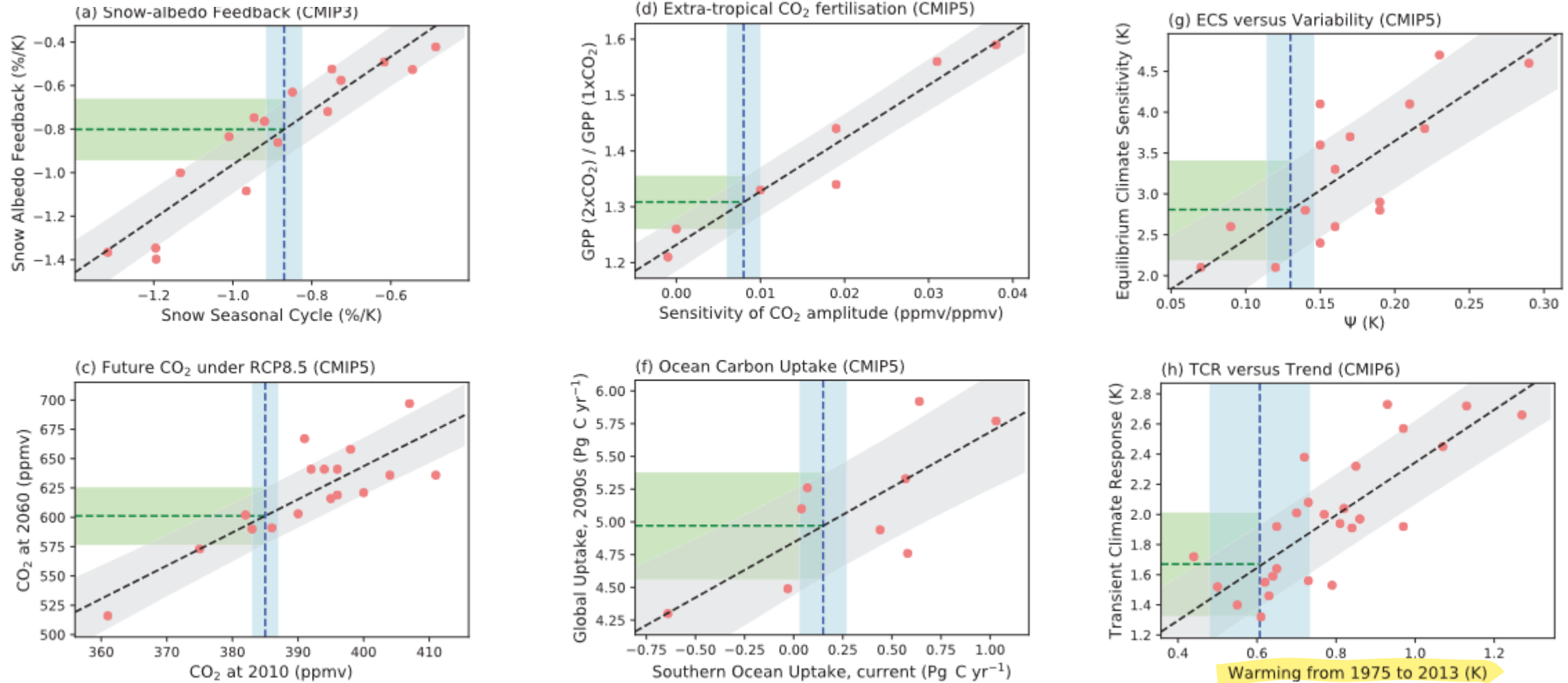
Emergent constraints

- The climate is a complex system
 - Only the basic physics is specified
- New relationships can be identified between properties of the system (they 'emerge')
 - Something we can observe (trend/variation)
 - Something we can't (a property of the Earth system)
- The snow-albedo feedback is an example of this
 - Seasonal variations in snow cover tell us about the response of snow to climate change



Williamson et al., Rev. M. Phys., 2021

Other examples



Detection and attribution

- Detection: *“demonstrating that climate has changed in some defined statistical sense, without providing a reason for that change”*
- Attribution: *“establishing the most likely causes for the detected change with some defined level of confidence”*
- As defined in the Third Assessment Report (IPCC, 2001)
- D&A is important for both weather and climate:
 - Attribution of extreme weather events to climate change (e.g., the 2018 heatwave)
 - Attribution of climate trends to human activities (are humans causing global warming?)

Fingerprints of human activities

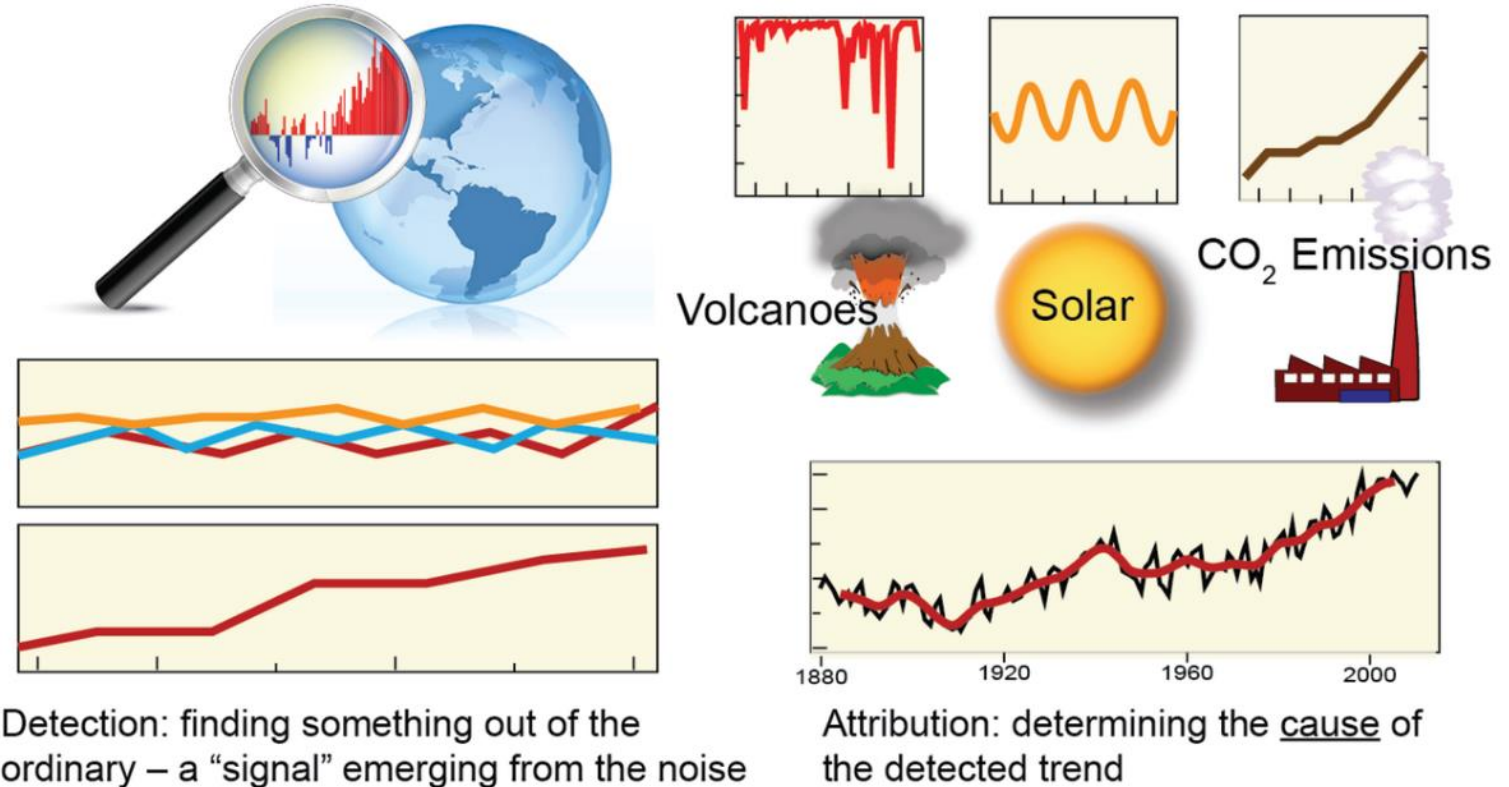
- Human-induced greenhouse gases leave specific **fingerprints** on climate, distinguishable from the effects of natural variability, volcanoes, the Sun, or other forcings
- These fingerprints are patterns in space or time
- They are important for attributing climate change to human activities
- Climate models help us determine what these fingerprints should look like
- The *pattern* of the fingerprint is more important than the magnitude of the signal: we know that climate models may get the magnitudes wrong, but they should get the correct pattern in space or time



Fingerprints of human activities

- The evolution of global-mean temperature matches the fingerprint of CO₂
- Volcanoes and solar activity have very different fingerprints in time

Detection and Attribution as Forensics



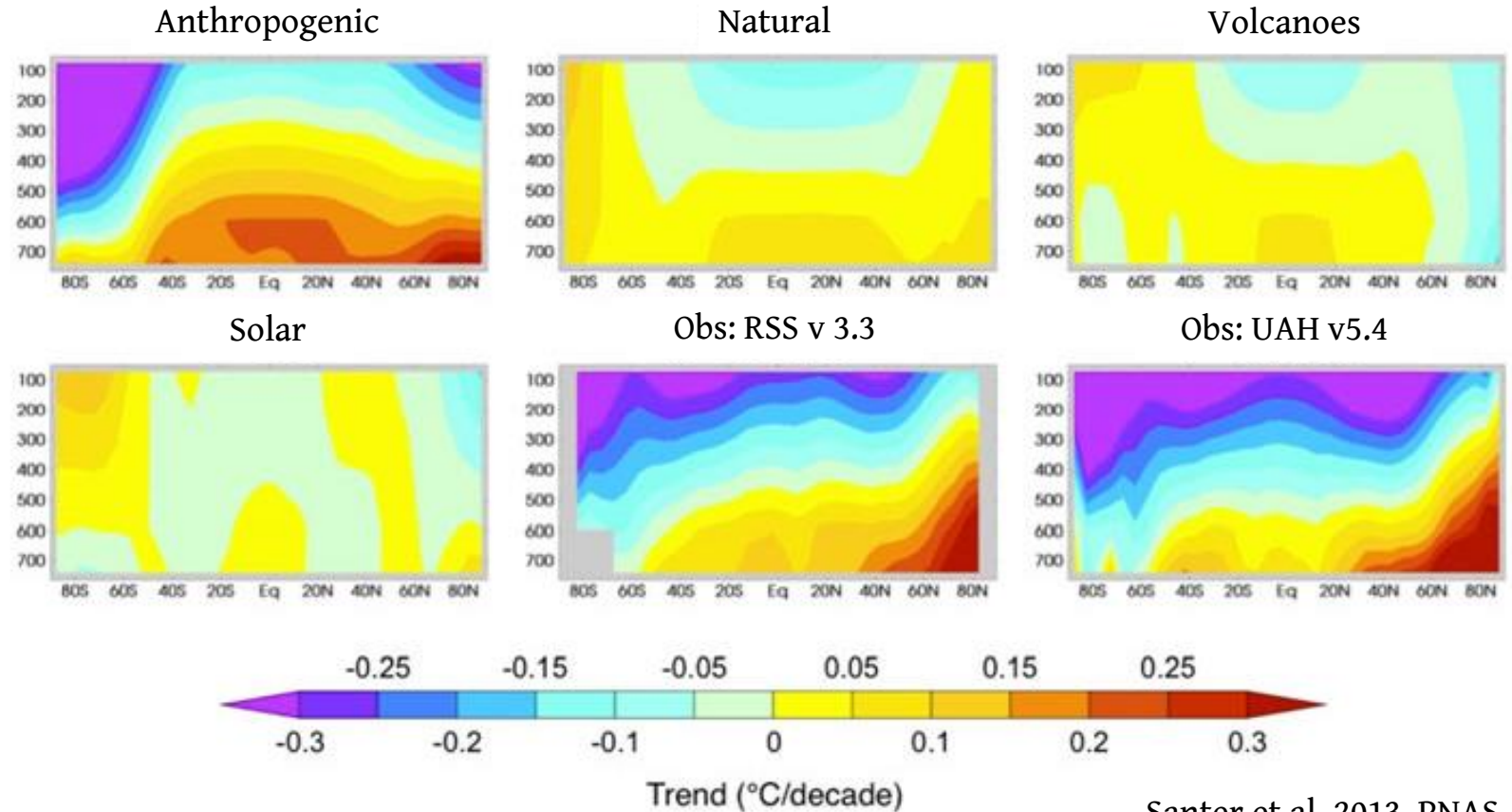
From Wikipedia: “Attribution of recent climate change”

Warming profile fingerprint

- GHGs also have a distinct fingerprint in the vertical warming profile
- Warming in the lower atmosphere but cooling in the upper atmosphere (stratosphere)
- Observations show this fingerprint

This shows warming because there was more volcanic forcing in the early part of the period

1979-2012 (33-year) temperature trends in CMIP5 and observations

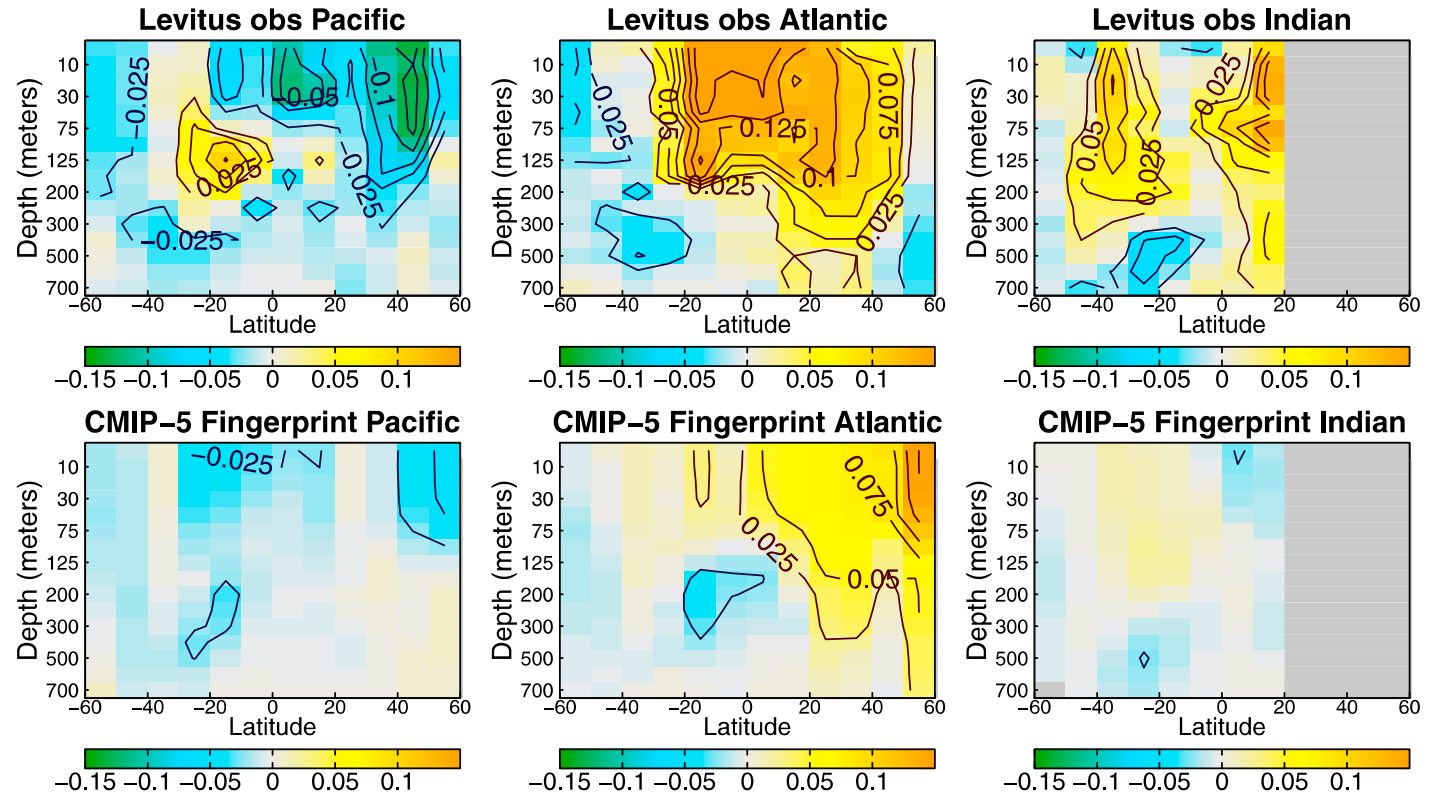


Santer et al. 2013, PNAS

Fingerprints in the ocean

- Human fingerprints also visible in ocean salinity
- Increase in the Atlantic, decrease in the Pacific
- This has to do with changes in evaporation and precipitation

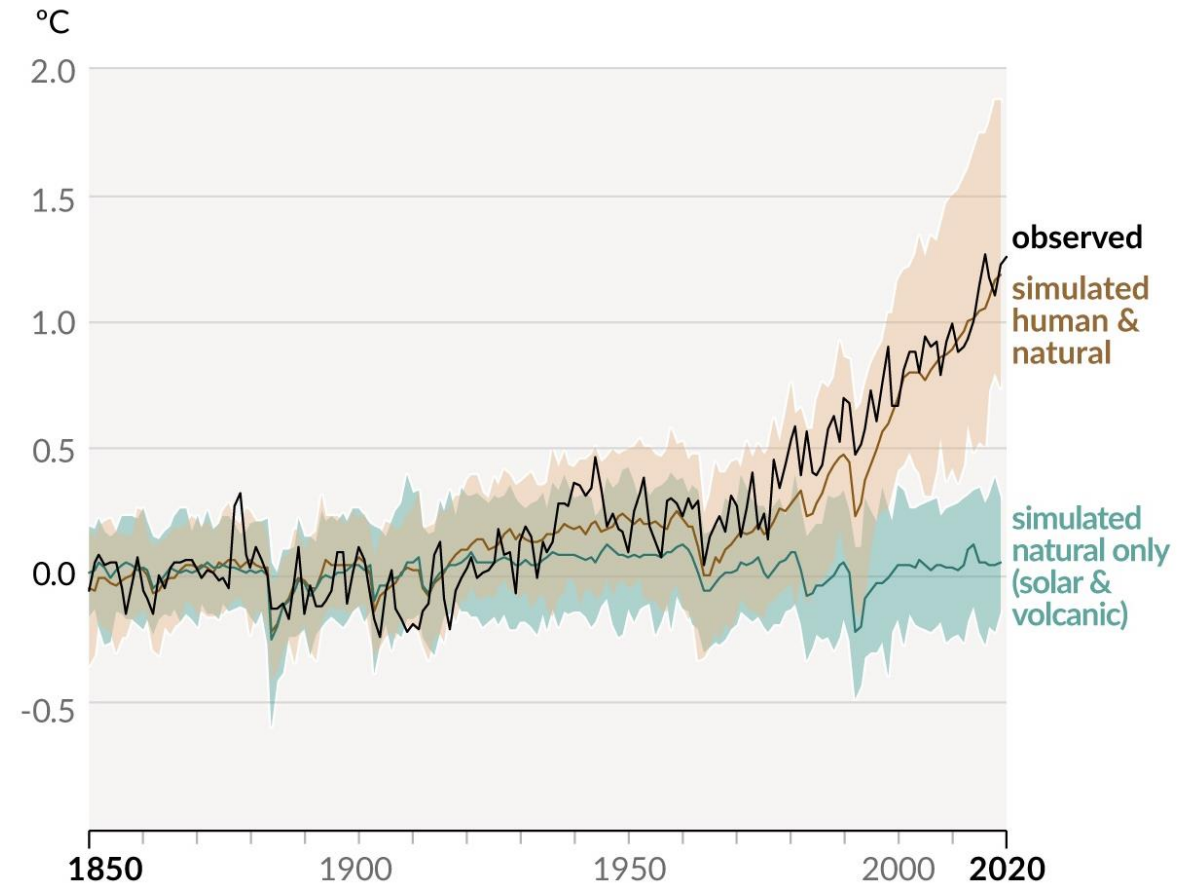
1955-2004 (50-year) salinity trends



Pierce et al. 2012, GRL

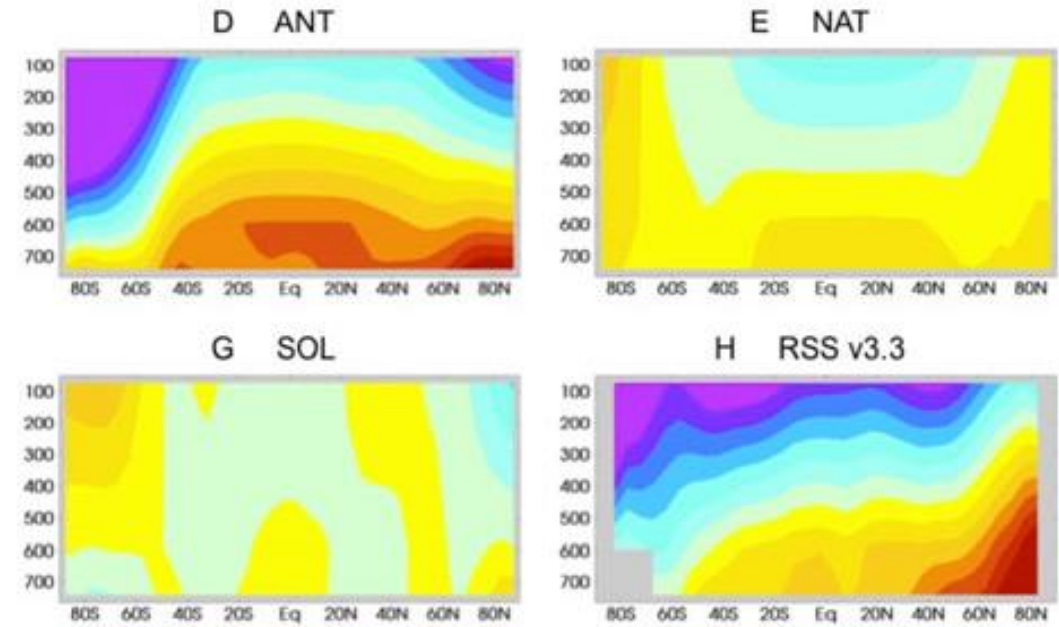
Attribution of climate trends

- Global warming since 1970s can only be explained by incorporating anthropogenic forcing (i.e. greenhouse gas emissions)
- This graph is the basis of the IPCC headline statement that human influence on the climate system is “unequivocal”



Summary: Evaluating climate models

- Climate models are far from perfect, but they remain our best tool to understand climate and project future changes
- All climate models are “tuned” to yield a realistic mean historical climate
- It may be possible to constrain climate projections using observable relationships, but this is still a subject of research
- Models are also used to identify “fingerprints” of human influence on climate



Questions

- How do you include small-scale processes in a climate model?
 - These processes are simplified (**parametrised**). This is a significant source of model uncertainty. What is the “equation for a cloud”?
- What constraints can we place on climate model behaviour?
 - We can match models to observations (e.g. of energy fluxes). **Emergent constraints** also allow us to understand behaviour under climate change
- How do we attribute climate change to different causes?
 - Models can uncover the “fingerprints” or warming from different sources.
 - Running models with only natural forcings can provide a ‘control’ Earth for comparison.

Further reading

- IPCC AR5 WG1 Chapter 9, section 9.2
- Blog post on climate model tuning:
<https://andthentheresphysics.wordpress.com/2017/09/05/climate-model-tuning/>